The Sharpe Stability Ratio

Temporal Consistency of Risk-Adjusted Performance

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ABSTRACT

This paper introduces the *Sharpe Stability Ratio* (*SSR*), a performance metric that evaluates the temporal consistency of risk-adjusted returns. While the Sharpe ratio (*SR*) summarizes average excess return per unit of risk over a fixed sample, it cannot distinguish persistent skill from episodic outperformance: two strategies may display identical ex-post SR yet differ sharply in their temporal profiles, with one delivering stable performance across subperiods and the other concentrating gains in a few favourable episodes. *SSR* addresses this gap by treating the rolling Sharpe ratio as a time-series object and defining stability as the ratio of mean rolling performance to its heteroskedasticity and autocorrelation consistent (HAC) standard deviation.

We show that this temporal-stability metric has several practical applications. First, *SSR* reveals that strategies with similar point-in-time *SR* and Probabilistic Sharpe Ratio (*PSR*) can exhibit markedly different stability profiles, providing information that is critical for due diligence and manager selection. Second, it quantifies the strong serial correlation induced by overlapping rolling windows and documents that naive dispersion measures severely understate uncertainty, implying that HAC correction is indispensable for valid temporal inference. Third, *SSR* supports formal hypothesis testing on stability via block bootstrap procedures that preserve the dependence structure of returns. Fourth, it demonstrates that statistically credible aggregate performance (*PSR* close to one) does not guarantee temporal consistency: high average *SR* may be generated by concentrated episodic strength rather than sustained skill. Evidence from controlled simulations and hedge fund index data shows that *SSR* delivers complementary insights relative to *SR* and *PSR*, separating genuinely stable performance from volatile profiles that appear credible under static evaluation.

JEL: G11, G12, C12, C14, C15

Keywords: Sharpe Stability Ratio, Sharpe ratio, Probabilistic Sharpe ratio, performance persistence, autocorrelation, HAC estimation, temporal stability, rolling Sharpe ratio, hedge fund performance, bootstrap inference.

1. INTRODUCTION

Risk–adjusted performance measures play a central role in portfolio selection, monitoring, and manager evaluation. Among them, the Sharpe ratio (SR) remains the dominant metric in both academic and applied settings, providing a scale-free summary of expected excess return per unit of risk with a well-known geometric interpretation as the tangency portfolio on the mean-variance frontier. Yet identical ex-post SR may conceal fundamentally different skill profiles: persistent alpha generation versus episodic outperformance interspersed with mediocrity. This distinction is economically material for monitoring, due diligence, and capital allocation.

Ex-post Sharpe ratios frequently mask temporal instability. A strategy with SR=1.2 calculated over three years may reflect genuinely consistent skill or the concentration of gains in a few favorable periods—standard inference cannot distinguish between these cases. Empirical evidence shows that much active fund outperformance reflects luck rather than skill (Fama and French, 2010), yet existing metrics evaluate only aggregate performance over a sample period. Traditional inference addresses whether observed performance is statistically significant but remains fundamentally pointwise in time: it summarizes performance over a fixed sample without regard to how that performance is distributed across subperiods.

Two strategies may exhibit identical full-sample Sharpe ratios yet differ substantially in their temporal structure. Calibrated performance may be broadly persistent across time or concentrated in a small number of episodes. Practitioners routinely compute rolling Sharpe ratios to visualize dynamics, yet no formal framework exists for statistical inference on temporal stability when these rolling estimates are constructed from overlapping windows that induce autocorrelation.

We address this gap through the *Sharpe Stability Ratio* (SSR), which answers: "*how stable is this Sharpe ratio over time?*". SSR is defined as the ratio of the mean rolling Sharpe ratio to its heteroskedasticity- and autocorrelation-consistent (HAC) standard deviation. Economically, SSR represents a *signal-to-noise* ratio for temporal performance consistency—high SSR indicates that strong risk-adjusted returns persist across subperiods, while low SSR reveals episodic outperformance. By construction, SSR penalizes low-frequency instability arising from serial dependence, regime persistence, or clustered realizations, while remaining agnostic to ex-post timing or reweighting decisions.

The key methodological challenge is that overlapping rolling windows induce strong serial correlation. For a given window length w and sample size T, adjacent windows share $w - 1$ observations, implying first-order autocorrelation close to one when w is large relative to the sampling frequency. Naive variance estimators ignore this dependence, understating uncertainty by orders of magnitude and invalidating classical inference. HAC correction is therefore essential for valid temporal inference about Sharpe ratio stability.

Our theoretical contributions are threefold. First, we quantify the autocorrelation structure of rolling Sharpe ratios under overlapping windows and demonstrate that conventional variance estimators are anti-conservative, overstating apparent temporal stability. Second, we establish the consistency of *SSR* under standard mixing conditions and derive its asymptotic distribution, showing that HAC estimation of the long-run variance delivers valid measures of temporal dispersion for stability inference. Third, we develop block bootstrap procedures tailored to preserve the dependence structure of rolling Sharpe ratios, obtaining finite-sample critical values and confidence intervals robust to strong serial correlation.

Methodologically, our framework differs from existing performance persistence studies that rank strategies by Sharpe ratios or alphas across non-overlapping periods, thereby discarding high-frequency temporal information. It also complements time-varying Sharpe ratio models, which focus on conditional expected returns and volatilities but do not deliver a scalar metric of unconditional temporal stability. The *SSR* framework:

- Preserves full temporal information by working with rolling windows rather than disjoint blocks.
- Corrects the induced autocorrelation of the rolling Sharpe sequence via HAC robust variance estimation.
- Provides formal hypothesis tests on temporal stability, ranging from absolute stability to threshold-based and pairwise comparisons across strategies.

Empirically, we demonstrate *SSR's* informational content through two complementary analyses. First, we construct synthetic strategies from different return-generating processes that yield similar ex-post Sharpe ratios but exhibit distinct temporal stability profiles. Second, we apply *SSR* to 17 BarclayHedge hedge fund strategy indices spanning 1997-2025. We show that *SSR* captures orthogonal information: strategies with similar *SR* can have substantially different SSR values, requiring distinct evaluation frameworks for due diligence and ongoing monitoring.

The remainder of the paper proceeds as follows. Section 2 reviews related literature positioning *SSR* relative to existing performance evaluation methods. Section 3 introduces the rolling Sharpe ratio, quantifies its autocorrelation structure from overlapping windows, develops HAC variance estimation, defines *SSR* formally, establishes its asymptotic properties, and analyzes its sensitivity to underlying parameters. Section 4 presents extensions: first, it positions *SR*, *PSR*, and *SSR* within a unified mean-variance framework across asset, credibility, and time spaces; second, it introduces the *Credibility-Adjusted SSR* (*CASSR*) combining point-in-time and temporal evaluation. Section 5 presents empirical applications to synthetic strategies and hedge fund indices, including robustness checks on window length, bandwidth selection, subperiod stability, and bootstrap inference. Section 6 concludes.

2. LITERATURE REVIEW

The evaluation of portfolio performance has evolved from simple return comparisons to sophisticated frameworks accounting for risk, statistical significance, and persistence. This section reviews the literature across four dimensions relevant to the Sharpe Stability Ratio: (1) point-in-time Sharpe ratio inference, (2) performance persistence and skill identification, (3) time-varying and conditional approaches, and (4) multiple testing and selection bias corrections. We identify the temporal stability gap that motivates *SSR*.

2.1 Classical Sharpe Ratio Inference

Roy (1952) pioneered the risk-reward framework through the *safety-first criterion*, maximizing the probability of exceeding minimum acceptable returns. Sharpe (1966) adapted this concept for mutual fund evaluation with the *reward-to-variability ratio,* measuring excess return per unit of total risk. The theoretical foundations linking this measure to mean-variance optimization and capital market theory were established earlier by Sharpe (1964), while Sharpe (1975) extended the framework to risk-adjusted performance measurement in practical portfolio contexts. Sharpe (1994) formalized the metric as the Sharpe ratio (*SR*), establishing the canonical formula, documenting statistical limitations, and defining best practices that elevated it to the universal performance standard.

Early statistical inference for the Sharpe ratio focused on hypothesis testing under restrictive assumptions such as IID normal returns (Jobson and Korkie, 1981; Memmel, 2003). Lo (2002) extended this framework by deriving the asymptotic distribution of the Sharpe ratio under serial correlation in returns, using heteroskedasticity- and autocorrelation-consistent (HAC) estimators. Mertens (2002) provides additional finite-sample corrections to Lo's framework. Opdyke (2007) further derives the asymptotic distribution of the Sharpe ratio under stationary and ergodic return processes, allowing for skewness, kurtosis, and other departures from IID assumptions. Ledoit and Wolf (2008) later proposed bootstrap-based robust procedures for testing differences in Sharpe ratios that improve finite-sample inference.

Bailey and López de Prado (2012) extended this framework through the *Probabilistic Sharpe Ratio* (PSR), which incorporates skewness and kurtosis into the sampling variance of the Sharpe ratio estimator, thereby accounting for departures from normality, answering "*Is this Sharpe statistically credible at this frequency over this sample?*".

These innovations establish a rigorous framework for point-in-time credibility assessment, yet they share a fundamental limitation: all evaluate performance at a single temporal aggregation without quantifying how risk-adjusted returns evolve within the sample period. A strategy may exhibit high *SR* and *PSR*—indicating statistically credible performance over the full sample—while this performance concentrates in brief episodic windows rather than persisting consistently across rolling subperiods. Existing metrics cannot distinguish genuinely stable alpha generation from volatile performance profiles that appear credible when aggregated but lack temporal consistency.

2.2 Performance Persistence and Skill Identification

A parallel literature examines whether superior past performance predicts future outperformance. Early studies (Grinblatt and Titman, 1992; Hendricks et al., 1993) documented short-term persistence in mutual fund returns. Carhart (1997) challenged these findings, demonstrating that apparent persistence largely disappears after controlling for common risk factors and transaction costs. Wermers (2000) reconciled these perspectives by distinguishing between gross and net returns, showing that fund managers exhibit stock-picking skill, but fees consume most value created. Bollen and Busse (2005) found short-term continuation patterns at quarterly frequencies that decay rapidly, while Fama and French (2010) concluded that most active mutual funds fail to deliver positive risk-adjusted returns net of costs.

Kosowski et al. (2006) introduced bootstrap methods to address the multiple testing problem inherent in evaluating thousands of funds simultaneously, showing that some funds in the extreme right tail exhibit genuine outperformance that cannot be attributed to luck.

However, these persistence studies share a common design: they rank strategies by past performance over non-overlapping periods and test whether rankings predict future performance. The focus is cross-sectional—sorting the universe to identify winners and losers—rather than characterizing within-strategy temporal profiles. A fund may rank in the top decile consistently (demonstrating persistence) while exhibiting high volatility in rolling Sharpe ratios within each period (indicating instability). *SSR* addresses this distinct dimension by quantifying temporal consistency for a given return stream, regardless of relative ranking.

2.3 Time-Varying and Conditional Approaches

An alternative perspective models the *SR* as time-varying, conditional on observable state variables or latent regimes. Tang and Whitelaw (2011) develop GARCH-based frameworks showing that expected risk-adjusted returns vary significantly with volatility regimes. Wong et al. (2012) construct the mixed Sharpe ratio to implement "bang-bang" optimal trading strategies where investment decisions depend on the sign of time-varying performance metrics. Kostakis et al. (2011) extend this using option-implied distributions for forward-looking market timing signals. Ang and Timmermann (2012) provide a comprehensive framework for dynamic performance evaluation through regime-switching specifications, while Patton (2006) contributes copula-based dynamic models that capture evolving dependencies between strategy returns and risk factors.

These conditional approaches require specification of parametric models linking returns to state variables. *SSR* complements these methods by providing an unconditional, modelfree stability statistic computed directly from realized return paths. The perspectives are complementary: conditional models explain sources of time-variation in performance, while *SSR* measures the magnitude of unconditional temporal instability.

2.4 Multiple Testing and Selection Bias

The proliferation of trading strategies has intensified concerns about data mining and selection bias. Harvey et al. (2016) demonstrate pervasive p-hacking in asset pricing research, arguing for dramatically higher t-statistic thresholds (>3.0) to account for multiple testing. Barras et al. (2010) apply false discovery rate (FDR) methods to mutual fund performance, showing that while most funds exhibit zero or negative alpha, a small minority demonstrate genuine skill even after FDR correction.

Bailey and López de Prado (2014) and Bailey et al. (2014) introduced the *Deflated Sharpe Ratio* (DSR) to address selection bias from multiple testing and backtest overfitting, adjusting for the number of trials conducted during strategy development—effectively asking "*Did I cherry-pick this strategy from many candidates?*". *DSR* deflates the observed Sharpe ratio based on independent strategy variations tested, expected maximum Sharpe ratio under the null, and correlation structure among trials. López de Prado et al. (2025) synthesized these advances into a unified framework for Sharpe ratio inference, providing systematic adjustments for estimation uncertainty, non-normality, and selection bias.

Table 1 summarizes the distinct dimensions of Sharpe ratio inference addressed by *PSR*, *DSR*, and *SSR* along three axes: the source of uncertainty each method targets, the null hypothesis each operationalizes, and the adjustment mechanism employed. Crucially, while *PSR* and *DSR* correct for statistical and combinatorial distortions in a single observed Sharpe ratio, neither addresses whether performance is *consistent over time*—the structural gap that *SSR* is designed to fill. The inferential architecture underlying the Sharpe ratio, *PSR*, and *SSR* is formally developed in Section 4.

TABLE 1
Different Sharpe Ratio Dimensions: *PSR*, *DSR* and *SSR*

Question	Metric	Dimension	Main Adjustment
What is the ex-post risk-adjusted return?	*SR*	Point-in-time level	Mean–variance trade-off (excess return per unit of volatility)
Is the Sharpe ratio statistically credible?	*PSR*	Point-in-time level	Higher moments, non-normality, effective sample size
Did I cherry-pick this strategy?	*DSR*	Cross-sectional selection bias / multiple testing	Number of trials, cross-strategy dependence
Is this Sharpe ratio stable over time?	*SSR*	Time-series consistency	HAC-robust temporal dispersion

An integrated evaluation framework combines all four metrics, though their optimal sequence depends on context.

In *strategy development with backtesting*, the pipeline proceeds as follows: (1) *PSR* assesses point-in-time credibility adjusted for higher moments and effective sample size, (2) *DSR* corrects for ex-ante multiple testing across N trials, and (3) SSR evaluates ex-post temporal stability of realized performance. *DSR* and *SSR* address distinct concerns: *DSR* adjusts for data mining during strategy discovery (cross-sectional across trials), while *SSR* measures performance persistence over time within each individual strategy (longitudinal dimension). A strategy may exhibit high *DSR* (survived rigorous testing) yet low *SSR* (episodic rather than persistent alpha), or conversely low *DSR* but high *SSR*, which justifies treating selection bias and temporal stability as complementary dimensions in a comprehensive evaluation.

By contrast, when *evaluating existing portfolios without proprietary backtesting* (e.g., mutual funds, hedge fund indices, published ETFs), *DSR* is less relevant—the evaluator performs no ex-ante selection over a large trial universe. In this context, the relevant framework uses *PSR* for point-in-time credibility and *SSR* for temporal consistency. Investors can favor strategies that jointly exhibit high PSR and high SSR, or apply Pareto-dominance criteria in (*SR*, *PSR*, *SSR*) space to identify non-dominated portfolios. This joint assessment is critical: strategies with similar *SR* and *PSR* may differ substantially in *SSR*, revealing whether strong performance reflects consistent skill or episodic outperformance concentrated in brief periods.

3. THE SHARPE STABILITY RATIO

3.1 Background and motivation

3.1.1 The Sharpe ratio

Let $\{R_t\}_{t=1}^{T}$ denote asset or portfolio returns observed, and $\{R_{f,t}\}_{t=1}^{T}$ the corresponding risk-free returns. Excess returns are

$$r_t = R_t - R_{f,t}, t = 1, \dots, T.$$

Assumption 3.1 *Stationarity and mixing*

The excess-return process $\{r_t\}$ is strictly stationary and α-mixing. This ensures consistency of HAC variance estimators and validity of block bootstrap procedures. The condition is satisfied by standard return models including AR, GARCH, and Markov-switching processes (see Appendix B for technical details).

The *Sharpe ratio* (*SR*) over the full sample [0,T] is

$$\widehat{SR}_T = \frac{\bar{r}}{\hat{\sigma}_{\mathrm{r}}},$$

where $\bar{r} = T^{-1} \sum_{t=1}^{T} r_t$ is the sample mean, and $\hat{\sigma}_r^2 = (T-1)^{-1} \sum_{t=1}^{T} (r_t - \bar{r})^2$ is the sample variance.

The *annualized Sharpe ratio* is $\widehat{SR}_T\sqrt{q}$, where q is the number of observations per year (frequency or annualization factor, with $q = 252$ for daily data, 52 for weekly, and 12 for monthly).

3.1.2 Probabilistic Sharpe Ratio

Bailey and López de Prado (2012) introduced the *Probabilistic Sharpe Ratio* (*PSR*) to assess point-in-time credibility of observed Sharpe ratios. Under the assumption that returns are stationary and ergodic (not necessarily IID), $\widehat{PSR}_T(SR^*)$ is the estimated probability that the true Sharpe exceeds a predefined benchmark SR^*, given the empirical distribution of returns, explicitly incorporating skewness, kurtosis, and sample length. Therefore, *PSR* answers: "*What is the probability that the true Sharpe ratio exceeds a benchmark* SR^*?".

$$\widehat{PSR}_T(SR^*) = P(\widehat{SR}_T > SR^*) \ = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{(\hat{\gamma}_4 - 1)\widehat{SR}^2}{4}}}\right)$$

where $\Phi(\cdot)$ is the standard normal CDF, γ_3 is skewness, γ_4 is kurtosis and T the sample length.

While, under ideal conditions (IID Normal returns and large samples), the *PSR* essentially preserves the ordering induced by the *SR*, in realistic finite-sample settings with non-Normal returns *PSR* introduces credibility adjustments that can differentiate, and potentially re-rank, portfolios with similar *SR* according to the statistical reliability of their performance estimates.

A natural question follows. "*How long a track record must be to reject the null hypothesis* H_0: $SR \leq SR^*$ *with a given confidence level?"*. Bailey and López de Prado (2012) formalize this through the concept of *Minimum Track Record Length* (*MinTRL*) (see Appendix A.1)

3.1.3 The Temporal Stability Gap

The literature reveals a fundamental gap: no existing framework provides statistical inference on temporal stability of risk-adjusted performance while properly accounting for autocorrelation induced by overlapping rolling windows.

1. First, practitioners routinely monitor rolling Sharpe ratios to visualize performance dynamics, yet these rolling statistics are almost universally presented without formal statistical inference. The high serial correlation in overlapping windows means naive sample dispersion severely understates true variability, making it impossible to distinguish genuine stability from noisy oscillation.

2. Second, existing persistence tests examine whether relative rankings persist across non-overlapping periods but cannot quantify within-strategy temporal stability. These are orthogonal dimensions requiring distinct measurement frameworks.
3. Third, time-varying Sharpe models provide conditional expected values but not unconditional temporal stability statistics. Both dimensions inform allocation decisions: conditional models guide timing, while unconditional stability metrics guide sizing and commitment horizon.
4. Finally, the heteroskedasticity and autocorrelation consistent **(**HAC) correction required for valid rolling window inference differs fundamentally from classical point-in-time Sharpe testing. Lo (2002) and Ledoit and Wolf (2008) apply HAC to correct for autocorrelation in the underlying return series when estimating a single Sharpe ratio, while *SSR* requires HAC to correct for autocorrelation in the rolling Sharpe ratio sequence itself—a second-order effect arising from window overlap.

The *Sharpe Stability Ratio* tries to address this gap by:

i. formalizing rolling *SR* as a time-series object with quantifiable autocorrelation structure,
ii. deriving HAC-consistent measures of long-run variability for the rolling *SR* process that account for overlap-induced dependence,
iii. establishing a *signal-to-noise performance ratio* that measures temporal consistency,
iv. and developing block bootstrap procedures that preserve dependence structure for finite-sample inference.

3.2. Construction of Sharpe Stability Ratio

3.2.1 Rolling performance process

For a given window length τ, define the rolling mean excess return and rolling sample variance as

$$\bar{r}_{t,\tau} = \frac{1}{\tau}\sum_{j=0}^{\tau-1} r_{t-j}\,, \qquad \hat{\sigma}^2_{t,\tau} = \frac{1}{\tau-1}\sum_{j=0}^{\tau-1}(r_{t-j} - \bar{r}_{t,\tau})^2, \qquad t = \tau, \dots, T$$

The corresponding rolling Sharpe ratio is

$$\widehat{SR}_{t,\tau} = \frac{\bar{r}_{t,\tau}}{\hat{\sigma}_{t,\tau}}, \qquad t = \tau, \dots, T.$$

Define the rolling performance process

$$\boldsymbol{Z} = \{Z_t\}_{t=1}^{N}\,, \qquad Z_t = \widehat{SR}_{t,\tau}, \quad N = T - \tau + 1$$

where $\boldsymbol{Z}$ is the time series of Sharpe ratios computed over N overlapping rolling windows of fixed length τ, using observations $\{r_{t-\tau+1}, \dots, r_t\}$. The index shift $t \mapsto t + \tau - 1$ maps the sequential index $t = 1, \ldots, N$ of the rolling process to the calendar index $\tau, \ldots, T$ of the underlying return series.

The choice of the rolling window length (τ) is intrinsically a smoothing decision, analogous to selecting a bandwidth in nonparametric estimation: shorter windows produce noisy, high-variance Sharpe estimates, whereas longer windows yield smoother but more "averaged-out" dynamics. Nevertheless, the window must contain a sufficient number of observations for the Central Limit Theorem to justify approximate normality of local Sharpe-type estimators.

3.2.2. Autocorrelation from Overlapping Windows

Unlike point-in-time $\widehat{SR}_T$, which aggregates all data into a single estimate, rolling *SR* sequence provides a time-varying profile of Sharpe ratio performance, allowing us to assess whether risk-adjusted performance remains stable or exhibits significant temporal variation, preserving full temporal information but at the cost of introducing serial dependence among consecutive estimates.

Z is built on consecutive rolling windows that share τ-1 observations, creating mechanical overlap that induces strong temporal dependence even when underlying returns are weakly dependent or serially uncorrelated. Consequently, Z exhibits high autocorrelation by construction, making pointwise dispersion measures inadequate for assessing temporal stability.

Proposition 3.1 *(Mechanical autocorrelation from overlapping windows)*

Suppose that $\{r_t\}$ are IID with finite second moments. For a fixed window length τ, linear rolling statistics such as rolling means exhibit first-order autocorrelation $\rho_1 \approx (\tau\text{-}1)/\tau$ as $T \to \infty$. More generally, for lags $k < \tau$, $\rho_k \approx 1\text{-}k/\tau$, while ρ_k is negligible for $k \geq \tau$ (derivation in Appendix A.2)**.**

Remark 3.1 *(Extension to the Rolling Sharpe Ratio via Delta Method)*

Proposition 3.1 is stated for rolling means, which are *linear functionals* of the underlying return series. The rolling Sharpe ratio $\widehat{SR}_{t,\tau}$ is a *nonlinear transformation* of two rolling statistics: the rolling mean $\bar{r}_{t,\tau}$ and the rolling standard deviation $\hat{\sigma}_{t,\tau}$.

Formally, let $g(x, y) = x/y$ and consider the first-order Taylor expansion of $g(\bar{r}_{t,\tau}, \hat{\sigma}_{t,\tau})$ around the population values (μ_r, σ_r). This yields

$$\widehat{SR}_{t,\tau} \approx \frac{\mu_r}{\sigma_r} + \frac{1}{\sigma_r}(\bar{r}_{t,\tau} - \mu_r) - \frac{\mu_r}{\sigma_r^2}(\hat{\sigma}_{t,\tau} - \sigma_r).$$

This linearization expresses the rolling *SR*, up to an additive constant, as a linear combination of deviations of the rolling mean and rolling standard deviation from their population values. The rolling variance $\hat{\sigma}_{t,\tau}^2$ is itself a rolling average of squared demeaned returns and therefore satisfies Proposition 3.1. Since the square-root mapping is continuously differentiable for $\sigma_r > 0$, a second application of the delta method transfers the same first-order autocorrelation structure from $\hat{\sigma}_{t,\tau}^2$ to $\hat{\sigma}_{t,\tau}$. Therefore, the rolling performance process $Z_t = \widehat{SR}_{t,\tau}$ inherits the approximate autocorrelation structure $\rho_k \approx 1 - k/\tau$ for lags $k < \tau$, with the approximation holding to first order in $1/\tau$ and higher-order remainder terms vanishing asymptotically as $\tau \to \infty$.

In finite samples, the delta-method approximation introduces higher-order terms that depend on the skewness of returns and the signal-to-noise ratio μ_r/σ_r. These terms are of smaller order and do not affect the leading autocorrelation structure. As shown empirically in Section 5.1, the rolling *SR* series exhibit autocorrelation patterns closely aligned with those of rolling mean returns across the three synthetic strategies, consistent with the first-order delta-method prediction. Figure 1 reports the autocorrelation function (ACF) of rolling mean returns, contrasting empirical estimates with the theoretical IID benchmark for a window length of $\tau = 36$.

FIGURE 1
ACF of Rolling Mean Returns: Empirical vs. Theoretical

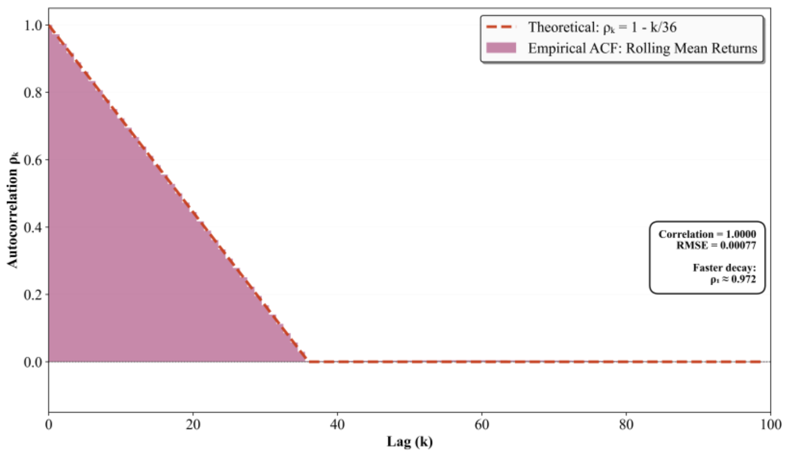


Figure 1 verifies Proposition 3.1 using simulated IID returns (T=10M, $\tau = 36$). Empirical ACF matches theoretical formula ρ_k=1-k/τ with correlation=1.

The *rolling performance process* inherits this autocorrelation structure. For τ=36 months, mechanical overlap predicts $\rho_1 \approx 0.97$, implying that consecutive observations are nearly perfectly correlated. This creates two critical problems:

1. *Visual smoothness is largely mechanical.* High autocorrelation in rolling *SR* plots primarily reflects overlapping window construction rather than genuine temporal stability.
2. *Naive variance estimators are severely biased.* Sample variance ignores autocorrelation, understating true uncertainty and invalidating classical *t*-tests for temporal inference.

Autocorrelation in $\boldsymbol{Z}$ thus arises from two sources: mechanical overlap (predictable from window length τ alone, present even with IID returns) and true serial dependence in underlying returns $\{r_t\}$. Detailed derivations and empirical evidence on the magnitude of variance understatement are provided in Appendix A.3 and the empirical analysis section 5.

3.2.3. Long-run variance and HAC estimation

The long–run variance of Z is defined as

$$\sigma_\infty^2(\boldsymbol{Z}) = \gamma_0 + 2\sum_{k\geq 1}\gamma_k\,, \qquad \gamma_k = \mathrm{Cov}(Z_t, Z_{t-k}).$$

whenever the sum converges.

Autocorrelation from overlapping windows invalidates 'naïve' sample variance as an estimator of uncertainty. Valid inference requires estimating the long-run variance, which accounts for serial correlation. We employ heteroskedasticity and autocorrelation consistent (HAC) estimation.

The Newey-West (1987) HAC estimator with Bartlett kernel is:

$$\hat{\sigma}_{\mathrm{HAC}}^2(\boldsymbol{Z}) = \hat{\gamma}_0 + 2\sum_{k=1}^{L}\kappa\left(\frac{k}{L+1}\right)\hat{\gamma}_k,$$

where $\hat{\gamma}_0$ is the sample variance, $\hat{\gamma}_k$ are sample autocovariances, $\kappa(z) = 1\text{-}|z|$ is the Bartlett kernel applying linearly declining weights to higher-order lags, and L is the bandwidth parameter.

We select bandwidth L using Andrews (1991) automatic data-dependent procedure, which adapts to the actual persistence structure of the rolling *SR* sequence. This ensures the HAC estimator accounts for both mechanical overlap and true serial dependence in returns. (Technical details in Appendix A.4)

3.2.4 The Sharpe Stability Ratio

Let $\mu_Z = E[\boldsymbol{Z}]$ denote the population mean of the rolling Sharpe process $\boldsymbol{Z}$, and let $\sigma_\infty(\boldsymbol{Z})$ denote its long-run standard deviation as defined in Section 3.2.3.

Structural Definition

The *Sharpe Stability Ratio* is defined as

$$SSR_0 = \frac{\mu_Z}{\sigma_\infty(\boldsymbol{Z})}.$$

The feasible sample analogue is

$$\widehat{SSR}_0 = \frac{\bar{Z}}{\hat{\sigma}_{HAC}(\boldsymbol{Z})}, \qquad \bar{Z} = \frac{1}{N}\sum_{t=1}^{N} Z_t.$$

Thus, SSR_0 is a signal-to-noise ratio for the rolling Sharpe process: it measures the magnitude of average risk-adjusted performance relative to its long-run temporal variability.

Inferential (Benchmark-Centered) Form

For hypothesis testing relative to a benchmark Sharpe ratio SR^*, define the centered statistic

$$SSR(SR^*) = \frac{\mu_Z - SR^*}{\sigma_\infty(\boldsymbol{Z})},$$

with sample analogue

$$\widehat{SSR}(SR^*) = \frac{\bar{Z} - SR^*}{\hat{\sigma}_{HAC}(\boldsymbol{Z})}.$$

Here, SR^*represents a required efficiency level, with $SR^* = 0$ serving as a natural baseline.

The uncentered quantity SSR_0 captures the structural persistence of performance, while the centered version $SSR(SR^*)$ provides the statistic used for formal inference.

3.2.5 Interpretation of the SSR

The Sharpe Stability Ratio (*SSR*) quantifies temporal performance consistency as a signal-to-noise statistic, accounting for autocorrelation in rolling Sharpe ratios. In its *inferential form* (benchmark-centered), the numerator measures excess performance over SR^*, while the HAC-adjusted denominator captures long-run variability of the rolling process. In its *structural form* (uncentered), the numerator reflects the unconditional mean performance.

Economic interpretation. SSR differentiates persistent skill from episodic luck. Two strategies with identical ex-post *SR* can display markedly different *SSR* values (see Section 5): a strategy with a stable rolling *SR* attains a high *SSR*, indicating consistent performance across subperiods, whereas a strategy whose rolling *SR* oscillates sharply attains a low *SSR*, suggesting that strong aggregate performance is driven by short-lived favourable episodes rather than sustained skill. High *SSR* values require both (i) superior average performance relative to benchmark, and (ii) low temporal dispersion of rolling Sharpe ratios.

Unlike the Sharpe ratio, which conflates the level and the stability of performance within a single full-sample statistic, the structural Sharpe Stability Ratio isolates the temporal dispersion of rolling performance, and thus explicitly separates average risk-adjusted performance from its temporal consistency. This distinction has direct implications for the evaluation of portfolio managers (disentangling skill from luck), portfolio construction (imposing stability constraints), and performance fee design (conditioning compensation on both a high *SR* and a high *SSR*).

3.3 Statistical Properties

3.3.1 Asymptotic Distribution

Under Assumption 3.1 and the regularity conditions stated in Assumptions B.1–B.2 in Appendix B, formal inference on $SSR(SR^*)$ proceeds via two complementary routes: asymptotic normality (Section 3.3.1) and block bootstrap (Section 3.3.2). The bootstrap is preferred in practice given the strong autocorrelation induced by overlapping windows; the asymptotic results provide the theoretical foundation. Section 3.3.3 formalizes the hypothesis testing framework common to both approaches.

Theorem 3.1 *(Consistency of HAC variance estimator)*

As $N \to \infty$ (equivalently $T \to \infty$ with fixed window length τ),

$$\hat{\sigma}^2_{\mathrm{HAC}}(\boldsymbol{Z}) \xrightarrow{p} \sigma^2_\infty(\boldsymbol{Z}).$$

Theorem 3.2 *(Central Limit Theorem for the rolling mean)*

The sample mean of the rolling performance process $\boldsymbol{Z}$ satisfies

$$\sqrt{N}(\bar{Z}-\mu_Z) \Rightarrow \mathcal{N}(0, \sigma^2_\infty(\boldsymbol{Z})).$$

The structural ratio SSR_0 is a population signal-to-noise object and does not depend on a reference level. For hypothesis testing relative to a required Sharpe ratio SR^*, the proper inferential statistic is the studentized quantity

$$t_{SSR}(SR^*) = \frac{\sqrt{N}\,(\bar{Z} - SR^*)}{\hat{\sigma}_{\mathrm{HAC}}(Z)} = \sqrt{N}\,\widehat{SSR}(SR^*)\,,$$

where we consider the centered quantity $(\mu_Z - SR^*)$, with sample analogue $(\bar{Z} - SR^*)$.

Theorem 3.3 *(Asymptotic normality of the SSR test statistic)*

Under the stated assumptions, and:

- *Under the boundary null* $H_0\colon \mu_Z = SR^*$,

$$t_{SSR}(SR^*) \Rightarrow \mathcal{N}(0,1).$$

- Under fixed alternatives $\mu_Z > SR^*$,

$$t_{SSR}(SR^*) \xrightarrow{p} +\infty.$$

Proofs are in Appendix B. The benchmark SR^*enters only through the null hypothesis; it does not affect the asymptotic variance, which is governed solely by the long-run variance of $\boldsymbol{Z}$.

3.3.2 Bootstrap Inference

Asymptotic approximations may be unreliable under the strong autocorrelation induced from overlapping windows. We therefore implement *moving block bootstrap* (MBB) procedures that preserve temporal dependence and provide finite-sample inference (Künsch, 1989; Hall et al., 1995).

The MBB resamples contiguous blocks of the original return series $\{r_t\}_{t=1}^{T}$, reconstructs bootstrap return paths, and recomputes $\widehat{SSR}(SR^*)$ for each replicate. For given block length ℓ:

1. Construct the set of overlapping blocks $B_j = (r_j, r_{j+1}, \dots, r_{j+\ell-1}),\ j = 1, \dots, T - \ell + 1$
2. Draw $K = \lceil T/\ell \rceil$ blocks with replacement, concatenate, and truncate to exactly T observations to obtain a bootstrap return path of length T.
3. Compute the rolling *SR* sequence and obtain $\widehat{SSR}^{(b)}(SR^*)$ for each bootstrap replicate $b = 1, \dots, B$.

Block length ℓ is selected via the Politis-White (2004) automatic procedure, which adapts to the autocorrelation structure each series. This yields the *empirical bootstrap distribution* $\{\widehat{SSR}^{(b)}(SR^*)\}_{b=1}^{B}$, from which confidence intervals and p-values are computed as described in Section 3.3.3. Detailed algorithms are in Appendix C.

3.3.3 Hypothesis Testing Framework

We formalize three complementary one-sided tests. In each case the p-value is computed from the bootstrap distribution of Section 3.3.2; under Theorems 3.1–3.3, the asymptotic $\mathcal{N}(0,1)$ critical values provide a large-sample alternative.

Test 1: *Mean rolling Performance vs. benchmark*

$$H_0: \mu_Z \leq SR^* \ \text{ vs. } \ H_1: \mu_Z > SR^*$$

The test statistic is $\widehat{SSR}(SR^*) = (\bar{Z} - SR^*)/\hat{\sigma}_{HAC}$, which measures how many HAC-adjusted standard deviations the mean rolling *SR* exceeds the benchmark. The bootstrap p-value is

$$\hat{p} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}\{\widehat{SSR}^{(b)}(SR^*) \leq 0\}$$

This percentile-method p-value is equivalent to checking whether the lower bound of the $(1-\alpha)$ bootstrap confidence interval for $SSR(SR^*)$ exceeds the null boundary: $\hat{p} < \alpha$ if and only if the α-quantile of $\{\widehat{SSR}^{(b)}(SR^*)\}_{b=1}^{B}$ lies above zero. The same equivalence applies to Tests 2 and 3 with their respective boundaries θ and 0. We reject H_0 at significance level α if $\hat{p} < \alpha$.

Test 2: *SSR exceeds a stability threshold θ*

$$H_0: SSR(SR^*) \leq \theta \quad \text{vs.} \quad H_1: SSR(SR^*) > \theta$$

where $\theta > 0$ is a pre-specified stability threshold distinct from SR^*. The test statistic is $\widehat{SSR}(SR^*) = (\bar{Z} - SR^*)/\hat{\sigma}_{HAC}(\boldsymbol{Z})$, evaluated against the threshold θ rather than against zero. The bootstrap p-value is

$$\hat{p} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}\{\widehat{SSR}^{(b)}(SR^*) \leq \theta\}$$

We reject H_0 at significance level α if $\hat{p} < \alpha$, concluding that the strategy's temporal stability significantly exceeds the threshold θ.

Test 3: *Pairwise stability comparisons*

Tests whether strategy A exhibits higher temporal stability than B:

$$H_0: SSR_A(SR^*) \leq SSR_B(SR^*) \quad \text{vs.} \quad H_1: SSR_A(SR^*) > SSR_B(SR^*)$$

A *joint block bootstrap* applied simultaneously to both return series preserves cross-sectional dependence. The bootstrap p-value is

$$\hat{p} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}\{\Delta\widehat{SSR}^{(b)} \leq 0\},$$

With bootstrap differences $\Delta\widehat{SSR}^{(b)} = \widehat{SSR}_A^{(b)}(SR^*) - \widehat{SSR}_B^{(b)}(SR^*)$. We reject H_0 at significance level α if $\hat{p} < \alpha$, concluding that strategy A exhibits significantly higher temporal stability than strategy B. Implementation details for all three tests are in Appendix C.

3.4 Sensitivity Analysis

SSR depends on three components: mean rolling performance ($\bar{Z}$), benchmark Sharpe ratio (SR^*), and temporal instability (σ_{HAC}). Understanding how *SSR* responds to changes in these components provides insight into the *economic channels* driving temporal stability.

3.4.1 Partial Derivatives

Treating $\bar{Z}$, SR^*, and σ_{HAC} as continuously varying, *SSR* admits simple partial derivatives in absolute terms:

1) *Sensitivity to average performance:*

$$\frac{\partial SSR}{\partial \bar{Z}} = \frac{1}{\sigma_{\text{HAC}}}$$

A marginal increase in $\bar{Z}$ raises *SSR* proportionally to the inverse of temporal dispersion. When instability is low, even modest performance improvements translate into large *SSR* gains—temporal stability acts as a *performance amplifier* (Figure 2).

FIGURE 2
Temporal Stability as a Performance Amplifier (*SSR* vs $\bar{Z}$)

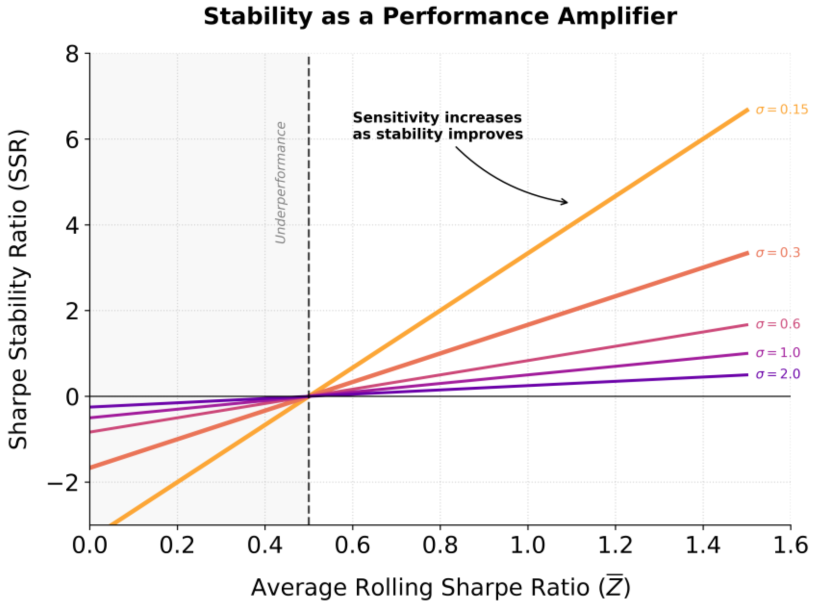


Figure 2. *Temporal Stability as a Performance Amplifier. The figure depicts the sensitivity of the SSR to changes in average performance $\bar{Z}$ across various levels of instability (SR^*=0.5). For fixed σ_{HAC}, the slope of each line corresponds to the partial derivative $\partial SSR/\partial \bar{Z} = 1/\sigma_{HAC}$, illustrating that lower HAC volatility enhances the marginal impact of performance gains. In the "Underperformance" region ($\bar{Z}$<SR^*), the SSR becomes negative regardless of stability. This relationship highlights the amplification property of temporal stability: strategies with lower σ_{HAC} exhibit a steeper response of SSR to changes in average performance compared to more volatile strategies.*

2) *Sensitivity to benchmark*

$$\frac{\partial SSR}{\partial SR^*} = -\frac{1}{\sigma_{\text{HAC}}}$$

Increasing the benchmark SR^* has equal magnitude but opposite sign to improving performance. The net effect on *SSR* depends on the *performance gap* $(\bar{Z} - SR^*)$.

3) *Sensitivity to temporal instability:*

$$\frac{\partial SSR}{\partial \sigma_{\text{HAC}}} = -\frac{\bar{Z} - SR^*}{\sigma_{\text{HAC}}^2} = -SSR\frac{1}{\sigma_{\text{HAC}}}$$

Reducing σ_{HAC} increases *SSR* nonlinearly: high-*SSR* strategies benefit more (in absolute terms) from stability improvements than low-*SSR* strategies.

3.4.2 Decomposition of SSR Changes

The total differential decomposes *SSR* changes into *three economic channels*:

$$dSSR \approx \frac{1}{\sigma_{\text{HAC}}}d\bar{Z} - \frac{1}{\sigma_{\text{HAC}}}dSR^* - \frac{SSR}{\sigma_{\text{HAC}}}d\sigma_{\text{HAC}}$$

1. *Performance channel* — improvements in average performance $(\bar{Z})$
2. *Exigence channel* — changes in benchmark (SR^*)
3. *Stability channel* — variations in temporal dispersion (σ_{HAC})

3.4.3 Elasticities

The elasticity of *SSR* with respect to temporal instability is particularly simple:

$$\varepsilon_{\sigma_{\text{HAC}}} = \frac{\partial SSR}{\partial \sigma_{\text{HAC}}} \cdot \frac{\sigma_{\text{HAC}}}{SSR} = -1$$

A 1% increase in σ_{HAC} produces a 1% decrease in *SSR*, regardless of $\bar{Z}$ or SR^*. *SSR* thus penalizes temporal instability in a *scale-invariant* manner.

By contrast, the elasticity with respect to average performance is:

$$\varepsilon_{\bar{Z}} = \frac{\partial SSR}{\partial \bar{Z}} \cdot \frac{\bar{Z}}{SSR} = \frac{\bar{Z}}{\bar{Z} - SR^*}$$

This diverges as $\bar{Z}$ approaches SR^*, indicating that small performance changes near the benchmark produce disproportionately large *SSR* effects.

The benchmark elasticity is:

$$\varepsilon_{SR^*} = \frac{\partial SSR}{\partial SR^*} \cdot \frac{SR^*}{SSR} = -\frac{SR^*}{\bar{Z} - SR^*}$$

Both performance and benchmark elasticities diverge as $\bar{Z} \to SR^*$, creating unstable SSR estimates near the benchmark boundary. This motivates the use of $SR^* = 0$ as a base benchmark in empirical applications: it eliminates divergence in economically relevant regions, as the elasticities become:

$$\varepsilon_{\bar{Z}} = 1, \qquad \varepsilon_{SR^*} = 0,$$

for strictly positive $\bar{Z}$.

The choice $SR^* = 0$ thus ensures that *SSR* sensitivity to performance changes remains stable and interpretable across the full range of realistic Sharpe ratios.

3.4.4 Iso-SSR Curves

For fixed SR^*, *SSR* level sets satisfy:

$$\bar{Z} = SR^* + \beta \cdot \sigma_{\mathrm{HAC}}$$

where β is the *SSR* level. These *iso-SSR curves* are upward-sloping straight lines in ($\bar{Z}$, σ_{HAC}) space with slope β (Figure 3). The slope measures the *marginal trade-off*: an increase in temporal instability must be compensated by β units of additional average performance to maintain constant *SSR*.

FIGURE 3

SSR Iso-curves: Performance-Stability Trade-off

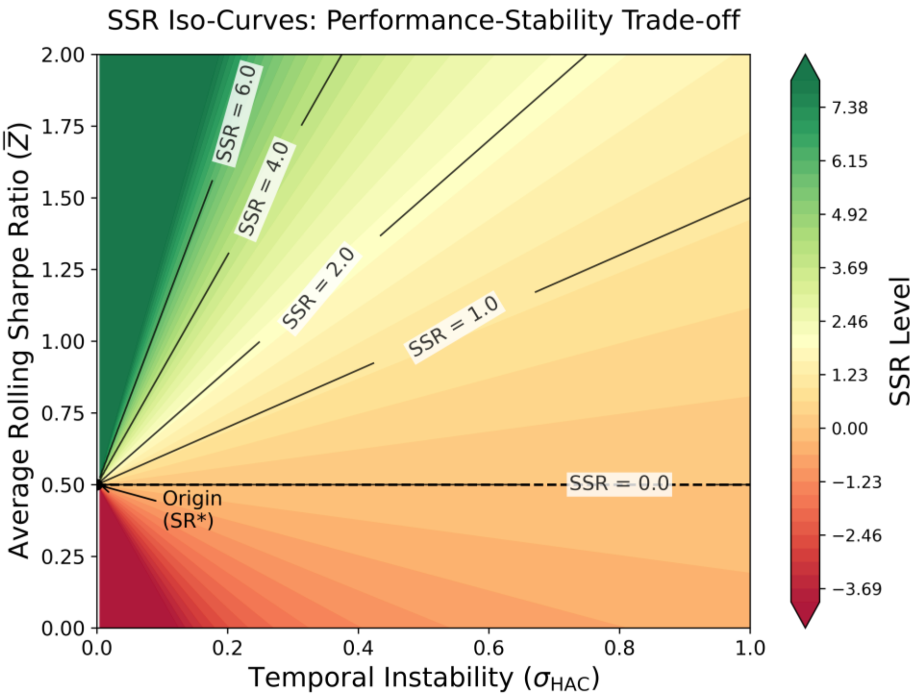


Figure 3. *Trade-off Map of the Sharpe Stability Ratio (SSR) with SR*=0.5. The contour plot illustrates the interaction between average performance ($\bar{Z}$) and temporal instability (σ_{HAC}). The colors represent the SSR level, while the black SSR-isocurves denote constant performance-stability combinations. The map reveals a linear trade-off: higher instability requires a proportionally higher mean Sharpe ratio to maintain a constant SSR.*

Key properties:

i. Higher SSR curves lie strictly above lower ones
ii. All curves emanate from the origin ($\sigma_{\text{HAC}} = 0$, $\bar{Z} = SR^*$)
iii. For fixed β, $\bar{Z} \to SR^*$ as $\sigma_{\text{HAC}} \to 0$. Slope β equals SSR level: strategies with higher SSR require proportionally more performance per unit instability

The map demonstrates that achieving higher temporal stability (lower σ_{HAC}) relaxes performance requirements: a stable strategy can attain the same SSR level with lower average Sharpe ratio than an unstable strategy.

4. STRUCTURAL FRAMEWORK

4.1 Structural and Inferential Architecture of Sharpe-Type Measures

Table 1 positioned *SR*, *PSR*, *DSR*, and *SSR* along four descriptive dimensions of performance evaluation. This section develops the structural foundations of three of these measures — *SR*, *PSR*, and *SSR* — leaving aside *DSR*, whose cross-sectional selection-bias correction operates independently of the temporal stability framework. We show that *SR* and SSR are not independent metrics, but structural signal-covariance objects defined over distinct domains — asset space and time space respectively — while *PSR* operates at a separate inferential layer applicable to either. This architecture motivates the *Credibility-Adjusted SSR* (CASSR) introduced in Section 4.2.

4.1.1 Sharpe Ratio: Asset-Space Structure

Let $R \in \mathbb{R}^n$ denote one-period gross returns on n risky assets with mean $\mu = E[R]$ and covariance $\Sigma = \text{Var}(R)$. For a risk-free rate r_f and portfolio weights $w \in \mathbb{R}^n$, define excess return $\mu^e = \mu - r_f \mathbf{1}$. The portfolio Sharpe ratio is:

$$SR(w) = \frac{w^\top \mu^e}{\sqrt{w^\top \Sigma w}}.$$

The mean–variance efficient frontier traces the upper envelope of expected excess return against portfolio volatility. Maximizing $SR(w)$ yields the tangency portfolio with optimal weights

$$w^\star \propto \Sigma^{-1} \mu^e,$$

and maximum Sharpe ratio

$$SR_{max} = \sqrt{(\mu^e)^\top \Sigma^{-1} \mu^e}.$$

The Sharpe ratio therefore evaluates cross-sectional performance by comparing a linear return signal to a quadratic dispersion measure induced by the return covariance operator

Σ. It is a structural object in asset space, independent of sampling or temporal aggregation considerations.

4.1.2 Probabilistic Sharpe Ratio: Inferential Adjustment

Let $\widehat{SR}_T(w)$ denote the sample Sharpe ratio computed over T observations for portfolio w. For a benchmark SR^*, the Probabilistic Sharpe Ratio is defined as

$$\widehat{PSR}_T(w; SR^*) = \Phi\left(\frac{\widehat{SR}_T(w) - SR^*}{\hat{\sigma}_{SR}(w)}\right),$$

where $\hat{\sigma}_{SR}(w)$ is the standard error of the Sharpe estimator, incorporating skewness, kurtosis, and sample size. *PSR* evaluates whether an estimated Sharpe ratio is statistically distinguishable from a benchmark once finite-sample uncertainty is considered. The numerator represents the estimated performance signal relative to the benchmark, while the denominator reflects sampling variability arising from limited observations and non-normal return distributions. Under the asymptotic normal approximation,

$$\widehat{PSR}_T(w; SR^*) = \mathbb{P}\big(\widehat{SR}_T > SR^*\big).$$

The resulting standardized statistic is mapped through the normal distribution to obtain a credibility probability in [0,1]. Unlike *SR* and *SSR*, *PSR* does not define a structural signal–covariance object over a domain of allocation. Rather, it operates at an inferential layer, quantifying the statistical credibility of estimated Sharpe-type statistics. As such, *PSR* can be applied to either *SR* or *SSR* when inference about statistical significance is required.

4.1.3 Sharpe Stability Ratio: Temporal Signal–Covariance Structure

We now consider temporal aggregation of the rolling performance process $\boldsymbol{Z}$. Let

$$\boldsymbol{Z} = (Z_1, \dots, Z_N)^\top, \qquad \boldsymbol{m_Z} = \mathbb{E}[\boldsymbol{Z}] = \mu_Z \mathbf{1}_N \in \mathbb{R}^N,$$

where $\mu_Z = \mathbb{E}[Z_t] \in \mathbb{R}$ denotes the scalar population mean of the rolling Sharpe process and $\mathbf{1}_N \in \mathbb{R}^N$ is a vector of ones.

Let $\boldsymbol{\Omega} \in \mathbb{R}^{N\times N}$ denote the long-run covariance matrix of $\boldsymbol{Z}$, consistent with the long-run variance defined in Section 3.2.3, reflecting serial dependence and mechanical overlap induced by rolling-window construction. For any temporal weight vector $\boldsymbol{a} \in \mathbb{R}^N$, define the aggregated performance score

$$S(\boldsymbol{a}) = \boldsymbol{a}^\top \boldsymbol{Z},$$

with mean $\boldsymbol{a}^\top \boldsymbol{m_Z}$ and temporal dispersion $\boldsymbol{a}^\top \boldsymbol{\Omega} \boldsymbol{a}$. If $\boldsymbol{\Omega} \succ 0$ and $\boldsymbol{m_Z} \neq 0$, the Rayleigh-quotient representation implies

$$\max_{a \neq 0} \frac{\boldsymbol{a}^\top \boldsymbol{m}_Z}{\sqrt{\boldsymbol{a}^\top \boldsymbol{\Omega} \boldsymbol{a}}} = \sqrt{\boldsymbol{m}_Z^\top \boldsymbol{\Omega}^{-1} \boldsymbol{m}_Z}, \quad \boldsymbol{a}^\star \propto \boldsymbol{\Omega}^{-1} \boldsymbol{m}_Z.$$

This unrestricted maximum parallels the tangency portfolio in asset space: the optimal temporal weight vector $\boldsymbol{a}^\star$ tilts toward subperiods with high expected rolling *SR* and low temporal covariance, analogously to how $w^\star \propto \Sigma^{-1}\mu^e$ tilts toward assets with high expected return and low covariance. However, $\boldsymbol{a}^\star$ implies an active timing strategy that overweights favorable subperiods ex post, which is economically inadmissible as an unconditional stability measure.

The *Sharpe Stability Ratio* is obtained by imposing the *no-timing restriction* $\boldsymbol{a} = \mathbf{1}_N$, which constrains all subperiods to receive equal weight. Since the sample mean is defined as $\bar{Z} = \frac{1}{N}\mathbf{1}_N^\top \boldsymbol{Z}$ — that is, the equal-weight normalization of the aggregated score — the long-run variance of the mean satisfies

$$\mathrm{Var}(\bar{Z}) = \frac{1}{N^2}\mathbf{1}_N^\top \boldsymbol{\Omega}\, \mathbf{1}_N \longrightarrow \frac{\sigma_\infty^2(\boldsymbol{Z})}{N} \text{ as } N \to \infty.$$

The structural Sharpe Stability Ratio is therefore defined as

$$SSR_0 = \frac{\mu_Z}{\sigma_\infty(\boldsymbol{Z})},$$

with feasible sample analogue $\widehat{SSR}_0 = \bar{Z}/\hat{\sigma}_{HAC}(\boldsymbol{Z})$.

The gap between the unrestricted maximum and the no-timing value of the functional measures the economic cost of the no-timing restriction:

$$\text{Timing gain} = \sqrt{\boldsymbol{m}_Z^\top \boldsymbol{\Omega}^{-1} \boldsymbol{m}_Z} - S(\mathbf{1}_N; \boldsymbol{m}_Z, \boldsymbol{\Omega}) \geq 0.$$

Using the asymptotic relation $S(1_N; m_Z, \Omega) \sim \sqrt{N}\, SSR_0$, this can be written as

$$\text{Timing gain} = \sqrt{\boldsymbol{m}_Z^\top \boldsymbol{\Omega}^{-1} \boldsymbol{m}_Z} - \sqrt{N}\, SSR_0.$$

Equality holds if and only if $\boldsymbol{m}_Z \propto \boldsymbol{\Omega}\mathbf{1}_N$, i.e., when the mean rolling *SR* vector is aligned with the covariance-weighted uniform direction — a condition approximately satisfied when temporal performance is homogeneous and $\boldsymbol{\Omega}$ is close to a scalar multiple of the identity.

In empirically relevant cases with regime-switching or clustered performance, the timing gain is positive, reflecting the improvement available to an investor who could identify and overweight high-*SR* subperiods in advance. SSR_0 deliberately abstracts from this possibility, delivering an unconditional stability measure that is agnostic to ex-post timing decisions.

4.1.4 Signal–Covariance Architecture and Inferential Layer

The Sharpe Ratio and the Sharpe Stability Ratio admit a common *signal–covariance representation*. Define the functional

$$\mathcal{S}(\boldsymbol{x};\boldsymbol{m},\boldsymbol{C}) = \frac{\boldsymbol{x}^\top \boldsymbol{m}}{\sqrt{\boldsymbol{x}^\top \boldsymbol{C}\boldsymbol{x}}}, \qquad \boldsymbol{C} \succ 0,$$

where $\boldsymbol{x}$ is a weight vector in a given domain, $\boldsymbol{m}$ is a signal vector, and $\boldsymbol{C}$ is a positive-definite covariance operator governing dispersion. The two instantiations of this functional differ in domain, weight vector, signal, and covariance operator:

- In ***asset space***, $\boldsymbol{x} = \boldsymbol{w} \in \mathbb{R}^n$, $\boldsymbol{m} = \boldsymbol{\mu}^e \in \mathbb{R}^n$, $\boldsymbol{C} = \boldsymbol{\Sigma} \in \mathbb{R}^{n\times n}$. The functional $\mathcal{S}(\mathbf{w};\boldsymbol{\mu}^e,\boldsymbol{\Sigma})$ equals the portfolio Sharpe ratio $SR(w)$. Its unconstrained maximum over $\boldsymbol{w}$ yields the tangency portfolio $\mathbf{w}^\star \propto \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e$with $SR_{\max} = \sqrt{(\boldsymbol{\mu}^e)^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}$.
- In ***time space***, $\boldsymbol{x} = \boldsymbol{a} \in \mathbb{R}^N$, $\boldsymbol{m} = \boldsymbol{m_Z} \in \mathbb{R}^N$, $\boldsymbol{C} = \boldsymbol{\Omega} \in \mathbb{R}^{N\times N}$. The functional $\mathcal{S}(\mathbf{a};\mathbf{m_Z},\boldsymbol{\Omega})$ is the temporal Rayleigh quotient. Its unconstrained maximum yields the optimal timing weight $\boldsymbol{a}^\star \propto \boldsymbol{\Omega}^{-1}\boldsymbol{m_Z}$ with maximum value $\sqrt{\boldsymbol{m_Z^\top}\boldsymbol{\Omega}^{-1}\boldsymbol{m_Z}}$.

A critical distinction separates the two domains. In *asset space*, the unconstrained optimizer $\boldsymbol{w}^\star$ is the object of interest: it is the efficient portfolio. In *time space*, the unconstrained optimizer $\boldsymbol{a}^\star$ is economically inadmissible as a stability measure, because it overweights subperiods with high rolling *SR* and low temporal covariance — an ex-post timing strategy that does not reflect unconditional performance consistency. The *Sharpe Stability Ratio* is therefore defined under the *no-timing restriction* $\boldsymbol{a} = \mathbf{1}_N$, which constrains all subperiods to receive equal weight:

$$SSR_0 = \lim_{N\to\infty} \frac{1}{\sqrt{N}}\, \mathcal{S}(\mathbf{1}_N;\mathbf{m_Z},\boldsymbol{\Omega}) = \frac{\mu_Z}{\sigma_\infty(\mathbf{Z})}$$

Thus SR and SSR_0 are both evaluations of the signal-covariance functional $\mathcal{S}$, but at different points: SR at the unconstrained optimum $\boldsymbol{w}^\star$ in asset space, and SSR_0 at the constrained equal-weight vector $\mathbf{1}_N$ in time space. The gap between the unrestricted temporal maximum $\sqrt{\mathbf{m_Z^\top}\boldsymbol{\Omega}^{-1}\mathbf{m_Z}}$ and SSR_0 measures the economic cost of the no-timing restriction, as developed in Section 4.1.3.

Statistical credibility operates at a distinct inferential layer. For any estimator $\hat{\mathcal{S}}$ of a signal–covariance functional, a standardized statistic

$$\mathcal{T} = \frac{\hat{S} - S^*}{\hat{\sigma}_S}$$

can be constructed, and the Probabilistic Sharpe Ratio applies the normal CDF to obtain a credibility probability. *PSR* therefore acts as an inferential transform of structural signal–covariance measures applicable to both SR and SSR_0.

4.2. Credibility-Adjusted SSR (CASSR)

4.2.1 Integrating Credibility and Temporal Stability

Within the layered signal–covariance framework developed in Section 4.1, *SSR* captures structural temporal persistence while *PSR* provides an inferential credibility transform reflecting sampling uncertainty in the estimated Sharpe ratio. *SSR* quantifies temporal stability of rolling Sharpe ratios but does not account for point-in-time statistical credibility—a strategy may exhibit high *SSR* (stable performance across windows) yet fail to achieve statistical significance when adjusted for skewness, kurtosis, and sample size, or conversely, demonstrate high *PSR* (credible point-in-time performance) but low temporal stability. This motivates the *Credibility-Adjusted Sharpe Stability Ratio* (CASSR), which combines inferential credibility (*PSR*) with temporal stability (*SSR*) to provide a unified metric assessing whether a strategy achieves both statistically credible performance and temporal consistency.

4.2.2 Definition of CASSR

In time space, taking $\boldsymbol{x} = \boldsymbol{a} \in \mathbb{R}^N$, $\boldsymbol{m} = \boldsymbol{m_Z} = \mu_Z \mathbf{1}_N \in \mathbb{R}^N$, and $\boldsymbol{C} = \boldsymbol{\Omega} \in \mathbb{R}^{N\times N}$ in the signal-covariance functional from Section 4.1 yields:

$$\mathcal{S}(\boldsymbol{a}; \boldsymbol{m_Z}, \boldsymbol{\Omega}) = \frac{\boldsymbol{a}^\top \boldsymbol{m_Z}}{\sqrt{\boldsymbol{a}^\top \boldsymbol{\Omega} \boldsymbol{a}}}.$$

Under the no-timing restriction $\boldsymbol{a} = \mathbf{1}_{\boldsymbol{N}}$, the numerator satisfies $\mathbf{1}_N^\top \boldsymbol{m_Z} = N\mu_Z$, so the functional evaluates to

$$S(\mathbf{1}_N; \boldsymbol{m_Z}, \boldsymbol{\Omega}) = \frac{N\mu_Z}{\sqrt{\mathbf{1}_N^\top \Omega \mathbf{1}_N}}.$$

Since the long-run variance of the sample mean satisfies

$$\mathrm{Var}\,(\bar{Z}) = \frac{1}{N^2} \mathbf{1}_N^\top \boldsymbol{\Omega} \mathbf{1}_N \longrightarrow \frac{\sigma_\infty^2(\boldsymbol{Z})}{N},$$

it follows that $\mathbf{1}_N^\top \boldsymbol{\Omega}\, \mathbf{1}_N \longrightarrow N\sigma_\infty^2(\mathbf{Z})$, and therefore

$$\mathcal{S}(\mathbf{1}_N; \boldsymbol{m_Z}, \boldsymbol{\Omega}) \longrightarrow \frac{N\mu_Z}{\sqrt{N}\,\sigma_\infty(\mathbf{Z})} = \sqrt{N} \cdot \frac{\mu_Z}{\sigma_\infty(\mathbf{Z})} = \sqrt{N} \cdot SSR_0.$$

Thus, under the no-timing restriction, the signal–covariance functional scales asymptotically as the studentized Sharpe Stability statistic at benchmark $SR^* = 0$,

$$t_{SSR}(0) = \frac{\sqrt{N}\,\bar{Z}}{\hat{\sigma}_{HAC}(\boldsymbol{Z})} = \sqrt{N}\,\widehat{SSR}_0,$$

as defined in Section 3.3.1.

The structural Sharpe Stability Ratio is therefore the per-$\sqrt{N}$ normalization of the Rayleigh functional in the limit,

$$SSR_0 = \lim_{N\to\infty} \frac{1}{\sqrt{N}}\, \mathcal{S}(\mathbf{1}_N; m_Z, \Omega),$$

with feasible sample analogue

$$\widehat{SSR}_0 = \frac{\bar{Z}}{\hat{\sigma}_{HAC}(\mathbf{Z})}.$$

For hypothesis testing relative to a benchmark Sharpe ratio SR^*, we consider the benchmark-centered statistic

$$\widehat{SSR}(SR^*) = \frac{\bar{Z} - SR^*}{\hat{\sigma}_{HAC}(\boldsymbol{Z})}.$$

The *Credibility-Adjusted Sharpe Stability Ratio* is defined as

$$CASSR(SR^*) = \widehat{SSR}(SR^*) \cdot \widehat{PSR}_T(SR^*) = \frac{\bar{Z} - SR^*}{\hat{\sigma}_{HAC}(\boldsymbol{Z})} \cdot \widehat{PSR}_T(SR^*).$$

where $\widehat{PSR}_T(SR^*)$ is the Probabilistic Sharpe Ratio computed over the full sample [1,T] as the inferential credibility transform applied to the estimated cross-sectional Sharpe ratio, and $\widehat{SSR}(SR^*)$ is the benchmark-centered Sharpe Stability Ratio defined in Section 3. CASSR operates entirely at the inferential layer, integrating temporal stability relative to a benchmark with statistical credibility of full-sample performance.

Note that $\widehat{PSR}_T(SR^*)$ is computed from the full-sample estimator $\widehat{SR}_T$, while $\widehat{SSR}(SR^*)$ is constructed from $\bar{Z}$, the mean of the rolling *SR* sequence. Although $\widehat{SR}_T$ and $\bar{Z}$ are distinct finite-sample statistics, both are consistent estimators of the same underlying population Sharpe ratio under Assumption 3.1, even though they need not coincide in finite samples. This is intentional: $\widehat{SR}_T$ is the efficient full-sample estimator that enters the credibility assessment of aggregate performance, while $\bar{Z}$ preserves the temporal structure of rolling estimates required for stability measurement. CASSR combines these two complementary perspectives — aggregate credibility and temporal consistency — rather than two competing estimators of the same object.

The *multiplicative form* is motivated by three considerations.

- First, since $\widehat{PSR}_T(SR^*) = \mathbb{P}(\widehat{SR}_T > SR^*)$ under the asymptotic normal approximation is a credibility probability, *CASSR admits a natural expected-value interpretation*: it represents a credibility-weighted temporal stability score, reflecting a probability-adjusted measure of stability conditional on the performance exceeding the benchmark:

$$\widehat{SSR}(SR^*) \cdot \widehat{PSR}_T(SR^*) \approx \mathbb{E}\left[\widehat{SSR}(SR^*) \cdot \mathbf{1}\{\widehat{SR}_T > SR^*\}\right]$$

  This approximation serves as an intuitive motivation rather than an exact identity: it would hold exactly if $\widehat{SSR}(SR^*)$ and $\mathbf{1}\{\widehat{SR}_T > SR^*\}$ were independent, a condition that fails in finite samples since both are functions of the same return series. The multiplicative structure is therefore justified on economic grounds, as formalized below.

- Second, *the multiplicative structure imposes a joint necessary condition*: *CASSR* vanishes whenever either dimension fails, reflecting the economic principle that neither credibility alone nor stability alone is sufficient for strategy acceptance. High *CASSR* values require both strong benchmark-adjusted temporal consistency and high point-in-time credibility. If credibility is low $\left(\widehat{PSR}_T \approx 0\right)$, *CASSR* converges to zero regardless of temporal stability; conversely, if benchmark-centered stability is weak $\left(\widehat{SSR}(SR^*) \approx 0\right)$, *CASSR* remains low even when credibility is high. In this way, *CASSR* provides a joint inferential measure that penalizes lack of either statistical reliability or temporal persistence.
- Third, since $\widehat{PSR}_T(SR^*) \in [0,1]$, multiplication operates as a credibility contraction of *SSR* toward zero without altering its scale or sign, whereas additive combinations would conflate metrics of fundamentally different magnitudes.

Changes in CASSR reflect variation in *seven performance drivers*:

1. *Mean Rolling Performance:* Changes in $\bar{Z}$
2. *Temporal Instability:* Changes in $\hat{\sigma}_{HAC}(\boldsymbol{Z})$
3. *Full-sample Performance:* Changes in $\widehat{SR}_T$
4. *Exigence:* Changes in benchmark SR^*
5. *Skewness Risk:* Changes in $\hat{\gamma}_3$
6. *Tail Risk:* Changes: $\hat{\gamma}_4$
7. *Track Record:* Changes in sample length T

This decomposition enables *performance attribution.* When comparing two strategies or analyzing temporal changes in $CASSR$, each channel's contribution can be quantified separately. This attribution should be understood as a comparative-static accounting decomposition rather than an orthogonal statistical decomposition.

While these effects are conceptually distinct, they are not empirically independent. In particular, $\bar{Z}$ and $\widehat{SR}_T$ are computed from the same return series and tend to be positively correlated, especially when the rolling window τ is large relative to T. Similarly, $\hat{\gamma}_3$ and $\hat{\gamma}_4$ typically covary with $\widehat{SR}_T$ under regime-switching dynamics or fat-tailed return distributions. In practice, changes in *CASSR* often reflect the joint movement of several channels simultaneously, and isolating individual contributions requires holding the remaining drivers fixed — a counterfactual exercise that can be informative for attribution purposes but does not imply statistical independence among channels.

4.2.3 When CASSR Matters

SSR is preferable when full-sample credibility is already established (*PSR* > 0.95) or when the analysis focuses exclusively on temporal consistency. It is particularly useful for comparing strategies with similar point-in-time credibility but different temporal stability profiles, or when working with long track records where statistical significance is not in question.

CASSR becomes essential when evaluating strategies with uncertain statistical significance (0.5 < *PSR* < 0.95), short track records, or returns exhibiting significant skewness or kurtosis. It provides a unified framework for simultaneous assessment of credibility and stability, making it suitable for comprehensive strategy ranking or due diligence processes requiring both dimensions.

Limitations. *CASSR* inherits assumptions from both components. The multiplicative form imposes symmetric proportional sensitivity between credibility and stability: halving *PSR* has the same impact as halving SSR. Investors with strong preferences for one dimension over the other may prefer alternative aggregators such as $\min\{PSR, SSR\}$ or weighted convex combinations, though these discard information about the magnitude of both dimensions simultaneously.

When $\bar{Z} < SR^*$, $\widehat{SSR}(SR^*)$ is negative so $CASSR$ is negative. In this region, $CASSR$ signals underperformance relative to the benchmark on both dimensions simultaneously; its magnitude reflects the degree of underperformance, preserving sign informativeness.

5. EMPIRICAL APPLICATION

In this section we examine the *SSR* under two complementary settings. Firstly, with synthetic strategies and secondly, with real data for 17 BarclayHedge hedge fund strategy indices over 1997–2025.

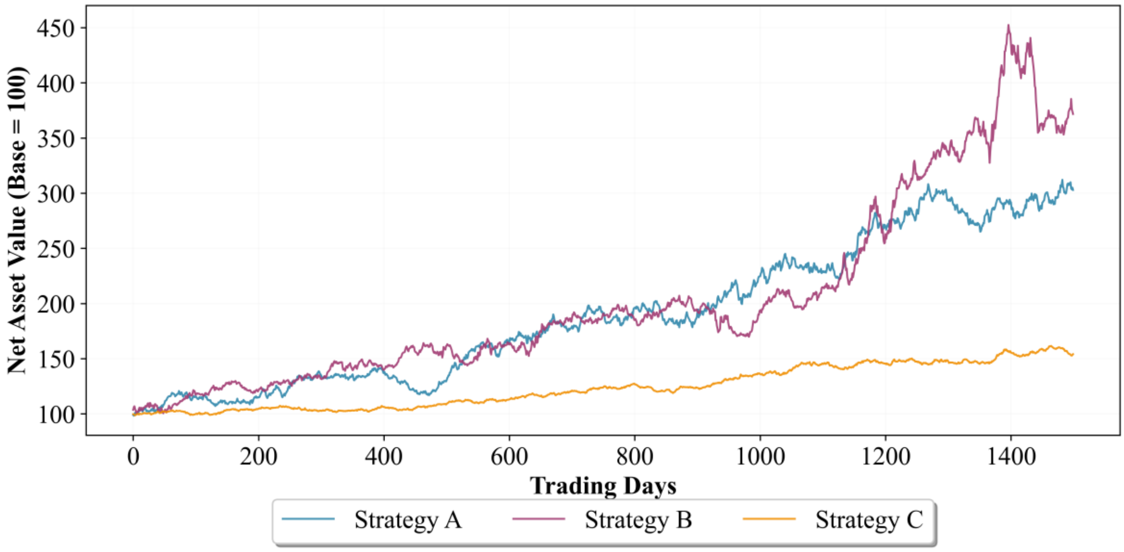
In principle, τ could be selected via data-driven criteria, such as minimizing an out-of-sample loss function in the spirit of rolling-window selection methods (e.g., Inoue et al., 2017). However, such procedures are model-dependent and computationally intensive, and no established optimality criterion exists for temporal stability measures such as *SSR*. Accordingly, we adopt conventional horizons widely used in performance monitoring: $\tau = 252$ trading days for daily data and $\tau = 36$ months for monthly data. These window lengths exceed the minimal asymptotic threshold commonly invoked for normal approximations. The selected horizons should therefore be interpreted as pragmatic and industry-standard rather than theoretically optimal choices. Sensitivity of *SSR* to alternative window lengths is examined in Section 5.3.1

5.1 Synthetic Strategies

We construct three synthetic strategies to isolate the informational content of *SSR* relative to static performance metrics. Real return data conflate multiple sources of heterogeneity (strategy changes, manager turnover, regime shifts) that make attribution of *SSR* differences to genuine temporal instability difficult. Synthetic strategies eliminate these confounds, enabling controlled measurement of the Sharpe ratio, its credibility (*PSR*), and its temporal stability (*SSR*).

The experiment is designed so that the three strategies exhibit nearly identical ex-post SR and PSR values, differing only in the temporal profile of returns. We generate daily return series (T = 1,500 observations, approximately six years) for three strategies with distinct return-generating processes. Strategy A follows an IID normal process (μ = 0.071% and σ = 1%). Strategy B employs a regime-switching model with transition probabilities p_{11} = p_{22} = 0.96, alternating between State 1 (μ = 0.049%, σ = 0.88%, unconditional probability 65%) and State 2 (μ = 0.20%, σ = 1.74%, unconditional probability 35%). Strategy C follows an IID normal process (μ = 0.03% and σ = 0.4%). Figure 4 displays cumulative performance.

FIGURE 4
Strategies' cumulative performance



For calculations, we use a rolling window length of $\tau = 252$ trading days and a benchmark Sharpe $SR^* \in \{0, 0.5, 1\}$.

TABLE 2
Statistics of Synthetic Strategies

Panel A. *Distributional Moments and Point-in-Time Performance*

Strategy	Mean (%)	StDev (%)	Skew	Kurt	Min (%)	Max (%)	*SR* (ann.)	*PSR* (0)	*PSR* (0.5)	*PSR* (1.0)
A	0.074	0.994	-0.016	3.23	-0.037	0.038	**1.185**	0.998	0.952	0.673
B	0.087	1.174	0.181	4.45	-0.042	0.059	**1.184**	0.998	0.953	0.674
C	0.029	0.386	0.013	3.11	-0.011	0.013	**1.184**	0.998	0.952	0.673

Panel B. *Sharpe Stability Ratio and temporal metrics*

Strategy	$\rho_1(\boldsymbol{Z_t})$	$\boldsymbol{\sigma_{Naive}}$	$\boldsymbol{\sigma}_{\text{HAC}}$	Bias ratio	$\bar{\boldsymbol{Z}}$	IQR $(\boldsymbol{Z_t})$	*SSR* (0)	*SSR* (0.5)	*SSR* (1)
A	0.988	0.576	2.990	0.193	1.199	0.742	**0.401**	0.234	0.067
B	0.994	0.792	4.251	0.186	1.232	1.117	**0.290**	0.172	0.055
C	0.993	0.746	3.985	0.187	1.281	1.151	**0.321**	0.196	0.071

Notes: All strategies constructed with T = 1,500 daily observations (~6 years). Mean return and StdDev (daily). Rolling window τ = 252 trading days, N = 1,249. HAC bandwidth L=30 using Andrews' (1991) automatic data-dependent procedure, $SR^* \in \{0, 0.5, 1\}$. IQR = q_{75}-q_{25}.

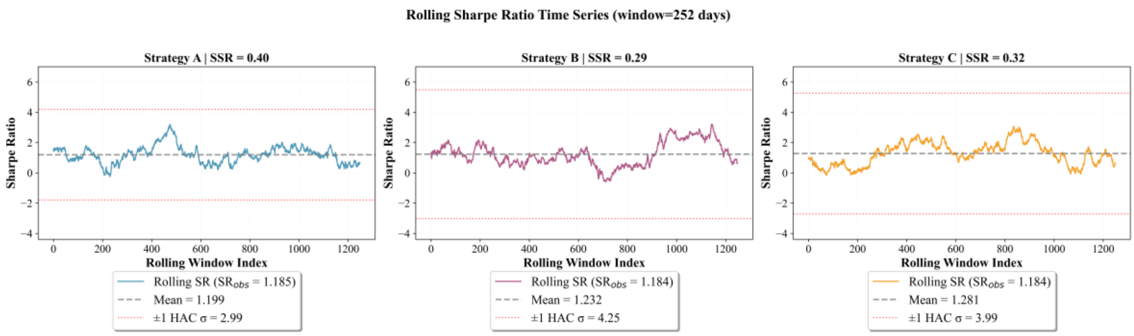
Table 2, **Panel A** shows all three strategies achieve statistically indistinguishable point-in-time performance: $SR \approx 1.18$, $PSR(0) \approx 0.998$, $PSR(0.5) \approx 0.95$, and $PSR(1.0) \approx 0.67$. Therefore, standard portfolio selection criteria based on these criteria would rank these strategies identically, implying their substitutability**.**

Panel B reveals this equivalence is misleading: temporal stability varies substantially across strategies despite identical aggregate performance. Strategy A achieves $SSR(0)$ = 0.40 versus 0.29 and 0.32 for Strategies B and C—a gap mainly driven by lower temporal dispersion (σ_{HAC} = 2.99 vs. 3.98-4.25). SSR decreases as SR^* increases, and its value changes according to the gap between the rolling mean SR ($\bar{Z}$) and the exigence (SR^*).

FIGURE 5

Rolling Sharpe Ratio dynamics

Panel A. *Evolution of* Z_t



Panel B. *Distribution of* Z_t

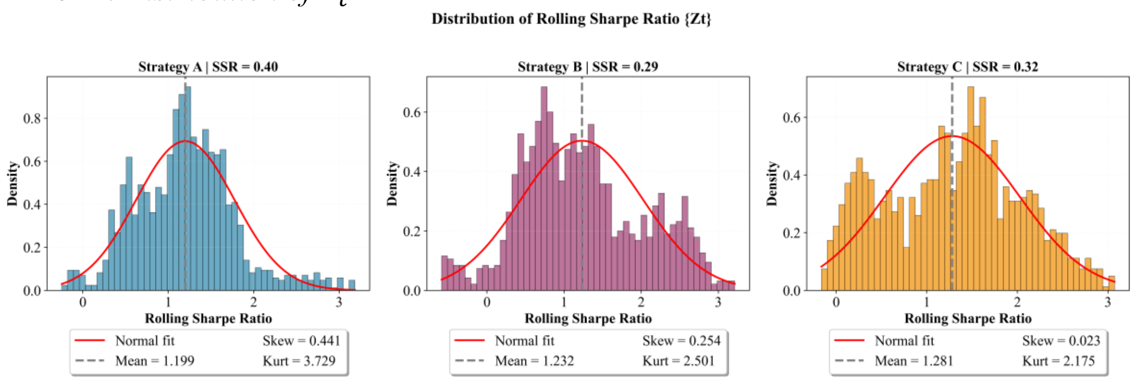


Figure 5, Panel A plots the time series evolution of rolling performance processes. These paths reveal that Strategy A displays a more stationary profile. **Panel B** plots the distributional differences. Strategy A exhibits a distribution more concentrated around its mean (IQR = 0.74), whilst Strategies B and C exhibit wider dispersion (IQR = 1.12 and 1.15, respectively).

FIGURE 6

Autocorrelation Structure of Rolling Returns and Rolling *SR*

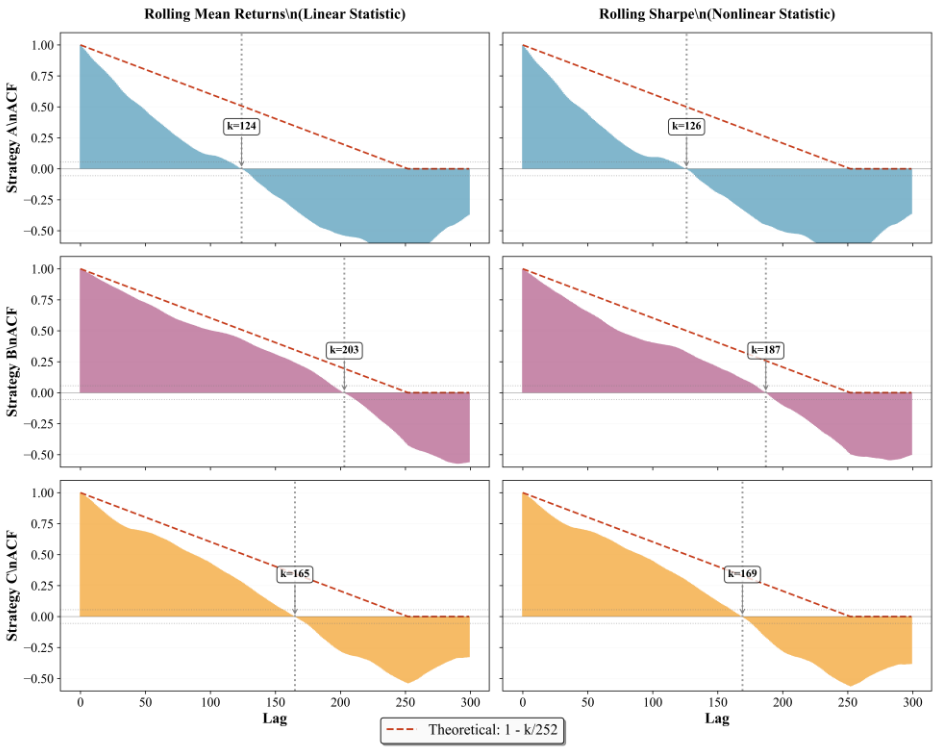


Figure 6 illustrates the autocorrelation structure of rolling statistics under mechanical overlap. All three strategies display rolling mean returns (left panels) with high autocorrelation coefficients, although below the theoretical $\rho_k \approx 1 - k/252$, reflecting finite-sample effects (T=1,500) and regime-induced serial dependence in Strategy B. Rolling *SR* (right panels) exhibits attenuated autocorrelation for Strategy B due to regime-induced serial dependence beyond mechanical overlap. Nevertheless, autocorrelation remains highly persistent for all strategies.

FIGURE 7

HAC vs. Naïve Standard Deviation of Rolling SR

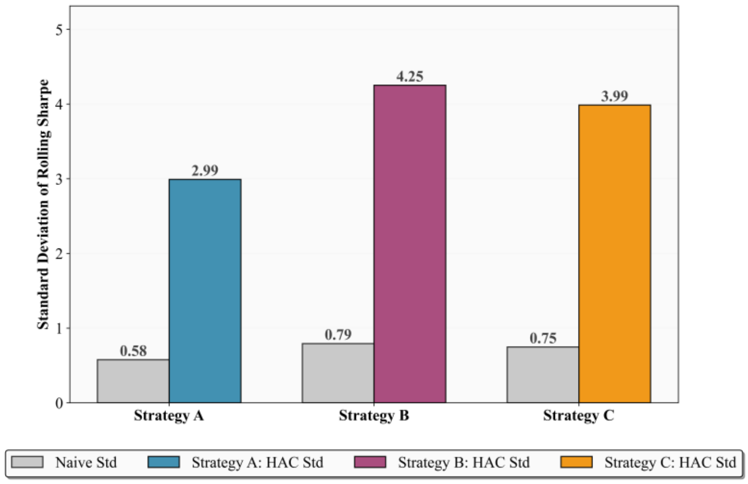


Figure 7 quantifies the HAC correction required for valid temporal stability inference. Naive standard deviations capture only ~19% of HAC-corrected estimates across all three strategies (bias ratios: Strategy A = 0.193, Strategy B = 0.186, Strategy C = 0.187), confirming that overlap-induced autocorrelation severely understates uncertainty regardless of the return-generating process.

This controlled experiment shows that temporal stability constitutes an independent dimension of performance evaluation: in cases where *SR* and *PSR* fail to distinguish among strategies, *SSR* reveals relevant differences in temporal consistency.

5.2 Hedge funds Strategies

Our second application examines temporal stability across 17 BarclayHedge hedge fund strategy indices from January 1997 to December 2025, yielding 348 monthly observations (29 years) per index. We use a 36-month rolling window ($\tau = 36$), producing $N = 313$ rolling *SR* observations per strategy.

This setting complements the synthetic portfolio experiments by assessing *SSR* on real-world strategies with pronounced non-normality, serial dependence and regime-switching behavior.

We adopt $SR^*= 0$ as the baseline benchmark for both *PSR* and *SSR* calculations, following Bailey and López de Prado (2012) and López de Prado et al. (2025). Since the Sharpe ratio is computed on excess returns, $SR^*= 0$ represents the null hypothesis of "no skill"—performance economically equivalent to holding cash[1]. We nonetheless compute *PSR* and *SSR* at higher thresholds (SR^*=0.5, 1.0) for robustness, recognizing that they represent economically meaningful hurdles for institutional-quality performance.

[1] This choice provides a conservative statistical baseline: rejecting H_0: SR $\leq$ 0 requires demonstrating that observed performance is incompatible with chance alone, even after adjusting for non-normality and serial correlation. While our BarclayHedge indices all exhibit full-sample *SR* > 0.6, this reflects survivorship and selection biases inherent in aggregated index data. Index-level aggregation systematically excludes underperforming funds, upward-biasing average performance. Nevertheless, SR^*=0 remains the theoretically grounded benchmark as it directly corresponds to the null distribution under the *PSR* and *SSR* inferential frameworks. Alternative benchmarks such as SR^*=0.5 or SR^*=1.0, while potentially reflecting investor expectations, lack theoretical grounding in the statistical framework and would arbitrarily exclude strategies with positive but modest risk-adjusted returns.

TABLE 3
Hedge Funds Performance Metrics

Panel A. *Distributional Moments and Point-in-Time Performance*

HF Strategy	Mean (%)	StDev (%)	Skew	Kurt	SR ann.	PSR (0)	PSR (0.5)	PSR (1)	MinTRL (0)	MinTRL (0.5)
Convertible Arbitrage	0.571	1.607	-2.659	26.582	1.229	1.000	0.991	0.772	5.00	14.06
Distressed Securities	0.601	1.935	-1.198	6.837	1.076	1.000	0.994	0.630	3.62	12.42
Emerging Markets	0.689	3.715	-0.719	7.009	0.642	0.999	0.759	0.039	7.86	158.51
Equity Long Bias	0.791	3.233	-0.610	4.827	0.847	1.000	0.956	0.228	4.63	27.13
Equity Long/Short	0.686	1.906	0.703	7.855	1.247	1.000	1.000	0.912	1.77	4.78
Equity Market Neutral	0.404	0.813	0.018	4.750	1.721	1.000	1.000	1.000	1.20	2.30
European Equities	0.704	2.165	1.503	11.316	1.127	1.000	1.000	0.780	1.75	5.48
Event Driven	0.658	2.031	-1.392	10.560	1.123	1.000	0.995	0.694	3.74	11.96
Fixed Income Arbitrage	0.459	1.254	-4.840	49.619	1.269	0.999	0.976	0.754	7.48	20.24
Fund of Funds	0.408	1.516	-0.851	7.822	0.932	1.000	0.977	0.377	4.29	19.65
Global Macro	0.583	1.588	0.615	4.310	1.272	1.000	1.000	0.940	1.56	4.10
Healthcare & Biotech.	1.127	4.667	1.911	18.605	0.837	1.000	0.979	0.162	3.16	19.08
Hedge Fund	0.655	2.015	-0.736	6.741	1.126	1.000	0.998	0.717	3.05	9.70
Merger Arbitrage	0.546	1.091	-1.419	14.861	1.732	1.000	1.000	0.993	2.41	4.67
Multi Strategy	0.579	1.299	-2.051	13.937	1.543	1.000	1.000	0.966	2.99	6.44
Pacific Rim Equities	0.668	2.783	0.264	7.766	0.832	1.000	0.960	0.187	4.13	25.47
Technology	0.952	3.738	0.492	6.210	0.882	1.000	0.982	0.259	3.42	17.84

***Notes*:** Mean return and StDev (monthly). Sharpe ratio annualized. Rolling window $\tau = 36$. T=348 months, N=313 windows. *MinTRL* (in years).

Table 3, Panel A reports distributional sample statistics and performance measures. *SR* range from 0.64 (*Emerging Markets*) to 1.73 (*Merger Arbitrage*), and all strategies display $PSR(0) \geq 0.999$, indicating statistically significant positive risk-adjusted returns at the aggregate level. Stricter benchmarks, however, reveal substantial heterogeneity in evidentiary strength: for $SR^*= 0.5$, 10 strategies attain $PSR(0.5) > 0.99$, whereas at $SR^*=1$ only 2 strategies achieve PSR(1) > 0.99.

Higher-moment characteristics show pronounced departures from normality that materially affect inference. Brooks and Kat (2002) report that the HFR Aggregate Hedge Fund returns index exhibits $\hat{\gamma}_3=-0.72$ and $\hat{\gamma}_4=5.78$. Despite the different time periods and databases, our *Hedge Fund index* yields similar results: $\hat{\gamma}_3=-0.73$ and $\hat{\gamma}_4=6.74$. Skewness ranges from -4.84 (*Fixed Income Arbitrage*) to +1.91 (*Healthcare & Biotechnology*), with 12 of 17 strategies exhibiting negative skewness, consistent with documented downside tail exposure in hedge fund returns. Kurtosis lies between 4.31 (*Global Macro*) and 49.62 (*Fixed Income Arbitrage*), far above the Gaussian benchmark of 3 and pointing to high tail risk, particularly among arbitrage-oriented styles.

These distributional features translate into markedly different evidentiary requirements when viewed through the MinTRL metric. For $SR^*=0$, *Emerging Markets* requires 7.9 years of data to confidently distinguish $SR>0$, despite posting an annualized Sharpe of 0.64, whereas *Equity Market Neutral* requires only 1.2 years—a roughly 6.5-fold difference driven by higher-moment risk rather than by the level of realized performance.

Panel B. *Sharpe Stability Ratio and temporal metrics*

HF Strategy	SR ann.	ρ1	σ_{Naive}	σ_{HAC}	Bias ratio	$\bar{Z}$	IQR (Z_t)	SSR (0)	SSR (0.5)	SSR (1)
Convertible Arbitrage	1.229	0.979	1.118	4.098	0.273	1.661	1.526	0.405	0.283	0.161
Distressed Securities	1.076	0.991	1.137	4.247	0.268	1.313	1.246	0.309	0.191	0.074
Emerging Markets	0.642	0.976	0.775	2.802	0.277	0.754	0.821	0.269	0.091	N/A
Equity Long Bias	0.847	0.956	0.602	2.055	0.293	0.869	0.723	0.423	0.179	N/A
Equity Long/Short	1.247	0.954	0.634	2.165	0.293	1.240	0.911	0.573	0.342	0.111
Equity Market Neutral	1.721	0.977	1.156	4.315	0.268	1.741	1.785	0.404	0.288	0.172
European Equities	1.127	0.968	0.647	2.321	0.279	1.143	1.000	0.492	0.277	0.062
Event Driven	1.123	0.969	0.717	2.540	0.282	1.183	0.960	0.466	0.269	0.072
Fixed Income Arbitrage	1.269	0.983	1.587	5.881	0.270	2.362	2.287	0.402	0.317	0.232
Fund of Funds	0.932	0.976	0.835	3.064	0.273	0.977	1.392	0.319	0.156	N/A
Global Macro	1.272	0.966	0.559	2.069	0.270	1.176	0.953	0.568	0.327	0.085
Healthcare & Biotech.	0.837	0.969	0.623	2.252	0.277	0.969	0.782	0.430	0.208	N/A
Hedge Fund	1.126	0.967	0.734	2.609	0.282	1.191	1.064	0.457	0.265	0.073
Merger Arbitrage	1.732	0.950	0.895	3.109	0.288	1.931	1.325	0.621	0.460	0.299
Multi Strategy	1.543	0.988	1.443	5.494	0.263	2.002	2.386	0.364	0.273	0.182
Pacific Rim Equities	0.832	0.965	0.680	2.350	0.289	0.888	1.014	0.378	0.165	N/A
Technology	0.882	0.956	0.586	2.025	0.290	0.920	0.743	0.454	0.207	N/A

Note: HAC bandwidth L=15 using Andrews (1991) automatic data-dependent procedure. IQR = q75-q25. $SR^*\in\{0, 0.5, 1\}$. Strategies reporting N/A for *SSR*(1) have full-sample *SR* below the benchmark SR^*=1, rendering the centered statistic economically uninformative at that threshold; inference for these strategies is conducted exclusively at $SR^*\in\{0, 0.5\}$.

Panel B reveals **substantial heterogeneity in temporal stability across strategies**, with *SSR*(0) values ranging from 0.27 (*Emerging Markets*) to 0.62 (*Merger Arbitrage*). Figure 8 plots the (*SR*,*SSR)* pairs for all strategies, while Figure 9 reports mean rolling performance against temporal volatility.

FIGURE 8

Sharpe ratio vs. *SSR*

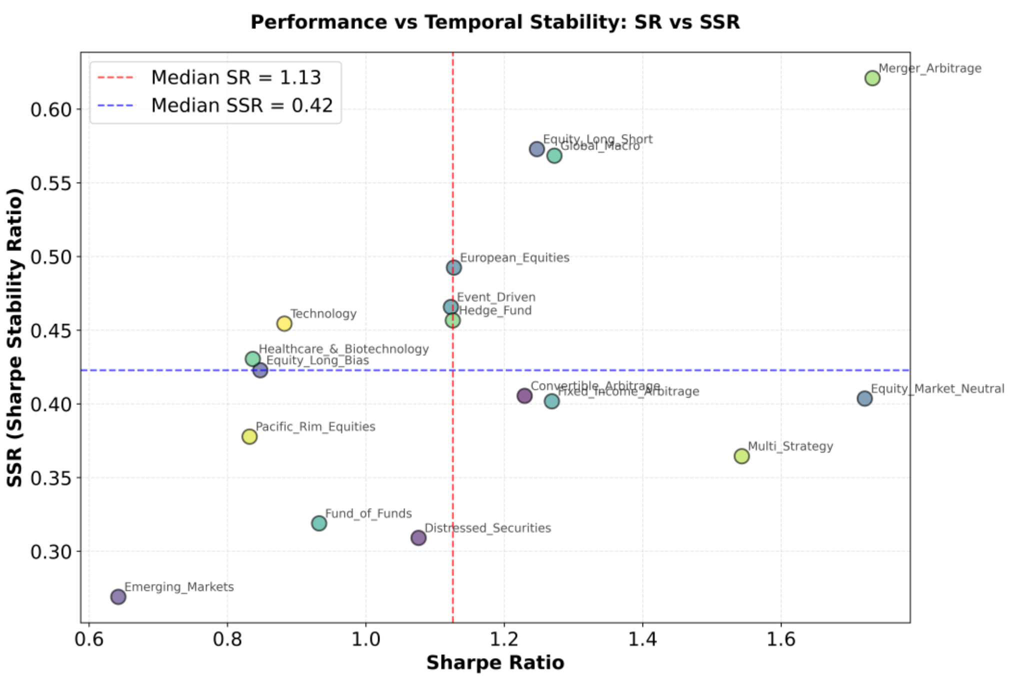


FIGURE 9

Mean Rolling performance vs. Temporal volatility

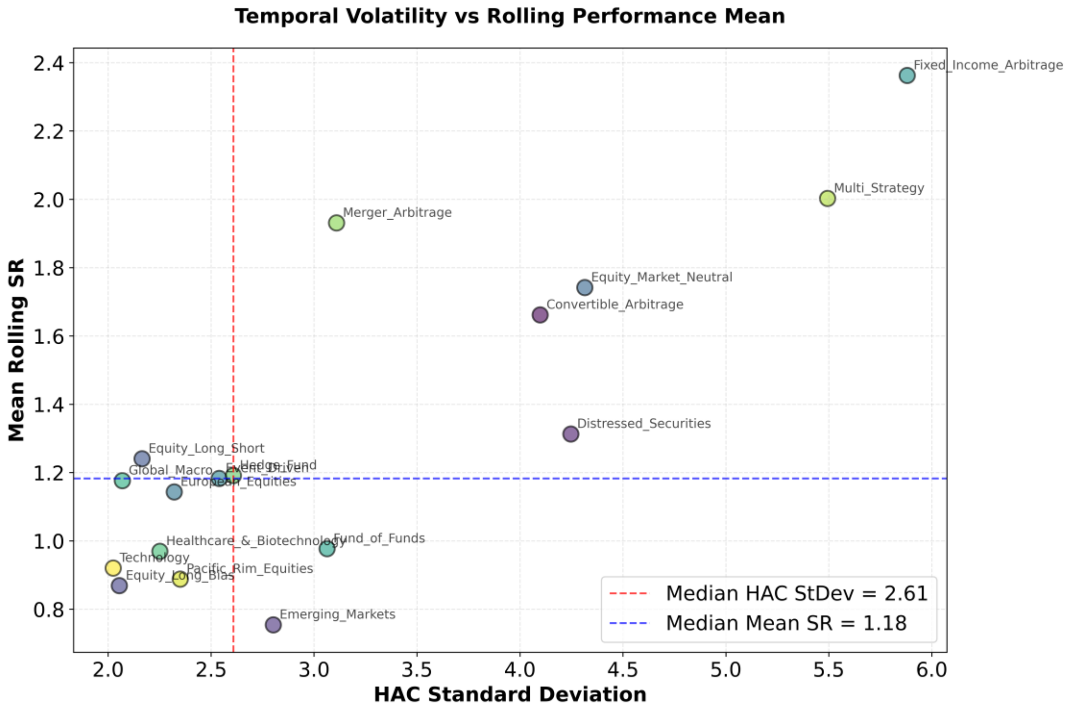


Table 4 presents the complete ranking shifts across *SR* and *SSR* for all 17 strategies and $SR^* = 0$. **Figure 10** shows the differences in ranking between both metrics.

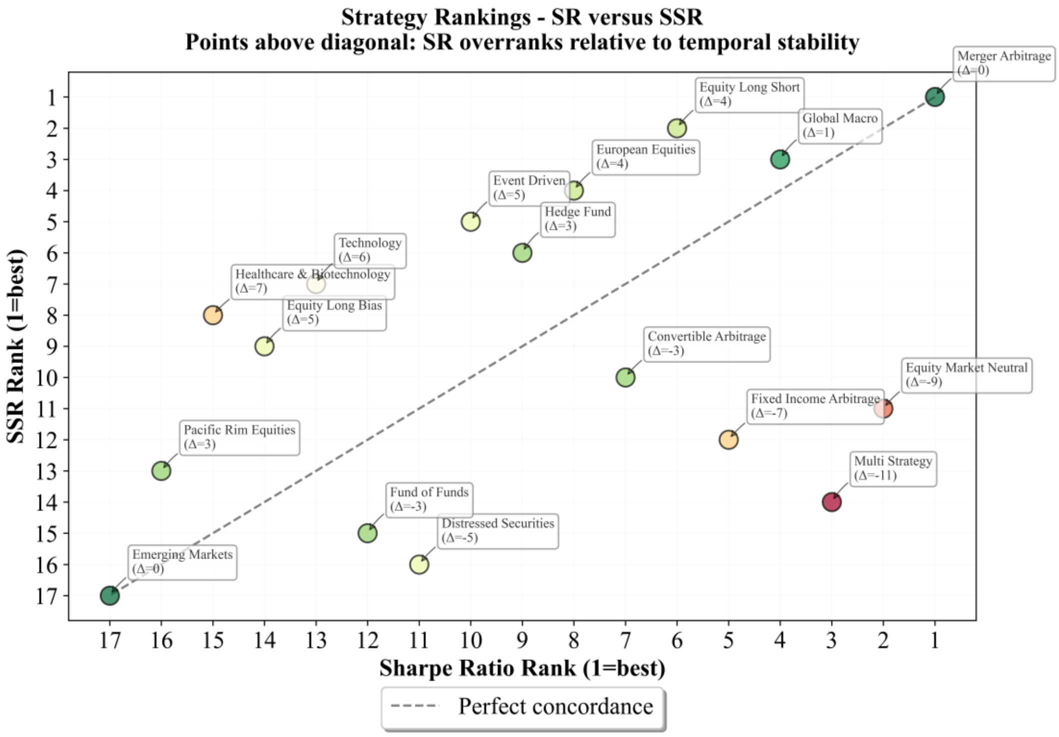
TABLE 4

Ranking Changes Across Performance Metrics ($SR^* = 0$)

HF Strategy	SR Rank	SSR Rank	Change (SR→SSR)
Merger_Arbitrage	1	1	*0*
Equity_Long_Short	6	2	*4*
Global_Macro	4	3	*1*
European_Equities	8	4	*4*
Event_Driven	10	5	*5*
Hedge_Fund	9	6	*3*
Technology	13	7	*6*
Healthcare_&_Biotechnology	15	8	*7*
Equity_Long_Bias	14	9	*5*
Convertible_Arbitrage	7	10	*-3*
Equity_Market_Neutral	2	11	*-9*
Fixed_Income_Arbitrage	5	12	*-7*
Pacific_Rim_Equities	16	13	*3*
Multi_Strategy	3	14	*-11*
Fund_of_Funds	12	15	*-3*
Distressed_Securities	11	16	*-5*
Emerging_Markets	17	17	*0*

Notes: *Rankings from 1 (highest) to 17 (lowest). Biggest improvements from SR to SSR: Healthcare & Biotechnology (+7), Technology (+6), Event Driven (+5), Equity Long Bias (+5), Equity Long/Short (+4),* *European Equities (+4). Biggest deteriorations: Multi Strategy (-11), Equity Market Neutral (-9), Fixed Income Arbitrage (-7), Distressed Securities (-5).*

FIGURE 10
Strategy Rankings: Sharpe ratio and *SSR* (SR^*=0)



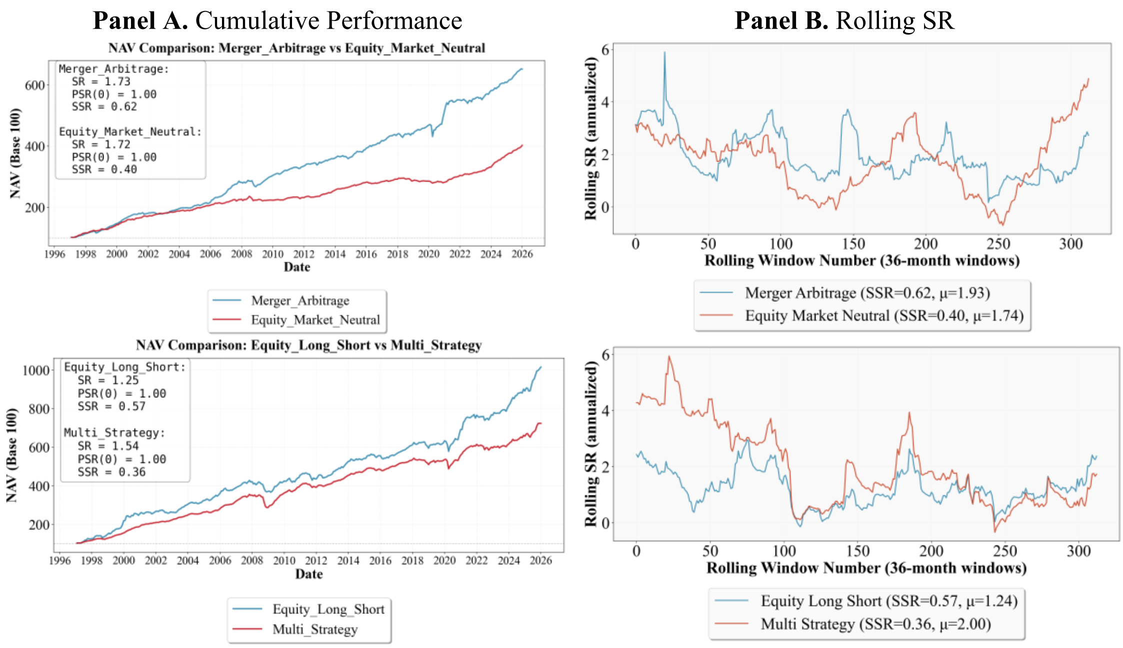
Comparing SR-based and SSR-based rankings reveals substantial reorderings that directly expose the distinction between static performance and temporal stability. High full-sample Sharpe ratios do not guarantee temporal consistency: strategies achieving top *SR* rankings can exhibit highly volatile rolling performance trajectories, characterized by intermittent periods of outperformance interspersed with extended underperformance. Conversely, strategies with modest static *SR* may demonstrate remarkably stable risk-adjusted returns over time, maintaining consistent performance across market regimes. This divergence reflects the fundamental difference between achieving a high average return-to-risk ratio and maintaining that ratio consistently.

Strategies suffering the largest *SSR* deteriorations share a common characteristic: elevated HAC standard deviations relative to their mean rolling *SR*, indicating that their attractive full-sample metrics mask significant temporal instability. Strategies gaining positions under *SSR* evaluation demonstrate the opposite pattern—stable rolling trajectories that deliver consistent risk-adjusted returns despite unremarkable averages. Only strategies simultaneously achieving high mean rolling *SR* and low temporal volatility maintain top rankings across both metrics, representing genuinely superior risk-adjusted performance.

Portfolio efficiency as measured by the Sharpe ratio or *PSR* does not exhibit a positive relationship with temporal stability. Figure 11 compares two strategy pairs to illustrate this decoupling. Panel A displays cumulative performance over 1997-2025, while Panel B presents the corresponding rolling *SR* trajectories.

FIGURE 11

Cumulative Performance and Rolling SR Comparisons for Strategy Pairs



The first pair illustrates how strategies with virtually identical static performance can display markedly different temporal stability. One strategy (*Merger Arbitrage*) and another (*Equity Market Neutral*) achieve nearly indistinguishable full-sample metrics (*SR* = 1.73 vs 1.72; *PSR*(0) = 1.00 for both), yet diverge sharply in *SSR*, at 0.62 versus 0.40 respectively. This 55% difference in stability arises entirely from their contrasting rolling *SR* dynamics: the former maintains HAC StDev = 3.11, whereas the latter exhibits substantially higher temporal volatility (HAC StDev = 4.31).

The second pair establishes that higher static performance does not guarantee superior temporal stability. *Multi Strategy* achieves *SR* = 1.54, exceeding *Equity Long/Short*'s *SR* = 1.25 by 24%, yet exhibits lower *SSR* (0.36 vs 0.57) due to higher temporal volatility (σ_{HAC} = 5.49 vs 2.17).

These comparisons of real strategies confirm that *SSR* captures dimensions of performance quality—specifically, the consistency of risk-adjusted returns through time—that remain invisible to point-in-time measures.

5.3 Robustness checks

To assess the stability of our *SSR* estimates, we conduct four complementary robustness analyses. First, we examine sensitivity to rolling window length, testing specifications from 24 to 60 months to verify that our main findings are not driven by the baseline 36-

month window choice. Second, we compare Newey-West and Andrews automatic bandwidth selection procedures to confirm that HAC variance estimation is robust to the lag truncation parameter. Third, we perform subperiod analysis across three distinct market regimes—pre-Global Financial Crisis (1997-2007), post-GFC (2008-2019), and post-COVID (2020-2025)—to verify temporal stability of *SSR* rankings and distinguish genuine skill persistence from regime-specific performance. Finally, we implement moving block bootstrap inference to construct confidence intervals on *SSR* estimates.

5.3.1 Window length sensitivity

Rolling window length (τ) represents a fundamental parameter choice in temporal stability analysis, involving a trade-off between statistical precision (larger windows) and responsiveness to regime changes (smaller windows). We evaluate *SSR* robustness across four specifications: $\tau = \{24, 36, 48, 60\}$ months, corresponding to 2, 3, 4, and 5-year windows respectively.

TABLE 5
SSR Robustness to Rolling Window Length

HF Strategy	τ =24	τ =36	τ =48	τ =60
Merger Arbitrage	0.43	0.62	0.74	0.85
Equity Long Short	0.45	0.57	0.67	0.73
Global Macro	0.52	0.57	0.60	0.61
European Equities	0.43	0.49	0.55	0.60
Event Driven	0.38	0.47	0.50	0.54
Hedge Fund	0.39	0.46	0.49	0.52
Technology	0.33	0.45	0.56	0.63
Healthcare & Biotechnology	0.34	0.43	0.50	0.59
Equity Long Bias	0.34	0.42	0.49	0.57
Convertible Arbitrage	0.39	0.41	0.41	0.43
Equity Market Neutral	0.38	0.40	0.43	0.48
Fixed Income Arbitrage	0.38	0.40	0.40	0.40
Pacific Rim Equities	0.31	0.38	0.43	0.47
Multi Strategy	0.35	0.36	0.37	0.37
Fund of Funds	0.30	0.32	0.33	0.33
Distressed Securities	0.28	0.31	0.32	0.35
Emerging Markets	0.24	0.27	0.29	0.32

Rank Correlation (Spearman): Adjacent: τ_{24}–τ_{36}: 0.83 | τ_{36}–τ_{48}: 0.96 | τ_{48}–τ_{60}: 0.99. Non-adjacent: τ_{24}–τ_{48}: 0.70 | τ_{24}–τ_{60}: 0.66 | τ_{36}–τ_{60}: 0.93 Average (all pairwise): 0.84

Notes: *SSR* calculated with HAC variance (L=15) for all window specifications. Baseline results use τ=36 months. Sample: BarclayHedge indices, January 1997-December 2025 (348 monthly observations). Rank correlations computed across all 17 strategies.

Table 5 reports *SSR* estimates across four window specifications. The results reveal strong rank stability despite systematic level effects. Spearman rank correlations between adjacent window lengths exceed 0.93, with an overall average of 0.84 across all pairwise comparisons, indicating that the relative ordering of strategies by temporal stability is robust to window choice.

Most strategies exhibit a monotonic upward trend in *SSR* as window length increases—for instance, Merger Arbitrage rises from 0.43 (τ = 24) to 0.85 (τ = 60), and Technology from 0.33 to 0.63. This pattern reflects increased statistical efficiency: longer windows reduce sampling variability in rolling Sharpe ratios, yielding more stable time series and hence higher signal-to-noise ratios. A few cases (such as *Fixed Income Arbitrage* or *Multi Strategy*) show almost flat *SSR* profiles across specifications, suggesting that for these strategies temporal stability is largely insensitive to reasonable changes in window length.

5.3.2 Bandwith Selection (Newey-West vs Andrews)

HAC variance estimation requires specifying the bandwidth parameter L, which determines how many autocorrelation lags to incorporate in the Bartlett kernel weighting. Two principal approaches are considered: the Newey-West (1987) rule-of-thumb formula L = floor(4(n/100)^(2/9)), and the Andrews (1991) data-dependent automatic selection procedure. To assess *SSR* sensitivity to this specification, we re-estimate all metrics under both bandwidth selection methods and compare results.

Table 6 presents *SSR* estimates using Andrews automatic bandwidth selection (baseline) versus the Newey-West rule. The Andrews procedure systematically selects L = 15 across all strategies, reflecting the strong autocorrelation present in rolling Sharpe ratio time series calculated over 36-month windows. In contrast, the Newey-West formula yields L = 5 for all strategies, a substantially shorter lag truncation driven by the mechanical formula's conservative parameterization for the sample size of rolling observations.

TABLE 6
SSR Robustness to Bandwidth Selection Method

Strategy	*SSR* (Andrews)	*L* (Andrews)	*SSR* (NW)	*L* (NW)	Δ SSR	Δ%
Merger Arbitrage	0.62	15	0.92	5	-0.303	-32.8
Equity Long Short	0.57	15	0.84	5	-0.269	-32.0
Global Macro	0.57	15	0.89	5	-0.319	-36.0
European Equities	0.49	15	0.75	5	-0.257	-34.3
Event Driven	0.47	15	0.70	5	-0.234	-33.5
Hedge Fund	0.46	15	0.69	5	-0.232	-33.7
Technology	0.45	15	0.67	5	-0.218	-32.4
Healthcare & Biotechnology	0.43	15	0.66	5	-0.228	-34.6
Equity Long Bias	0.42	15	0.62	5	-0.198	-31.9
Convertible Arbitrage	0.41	15	0.62	5	-0.219	-35.1
Equity Market Neutral	0.40	15	0.63	5	-0.226	-35.9
Fixed Income Arbitrage	0.40	15	0.62	5	-0.220	-35.3
Pacific Rim Equities	0.38	15	0.56	5	-0.180	-32.3
Multi Strategy	0.36	15	0.58	5	-0.211	-36.7
Fund of Funds	0.32	15	0.49	5	-0.173	-35.2
Distressed Securities	0.31	15	0.48	5	-0.171	-35.6
Emerging Markets	0.27	15	0.41	5	-0.141	-34.4

Spearman Rank Correlation: 0.98

Notes: SSR calculated with 36-month rolling window. Andrews bandwidth selected via data-dependent procedure. Newey-West bandwidth via formula L = floor(4(n/100)^(2/9)). ΔSSR = SSR(Andrews) - SSR(NW). Sample: BarclayHedge indices, January 1997-December 2025 (348 observations, ~313 rolling windows).

The results reveal large level differences but very stable rankings. SSR estimates under Andrews are on average 34% lower than under Newey–West (absolute gap ≈0.22), because the longer bandwidth $L = 15$ incorporates more autocorrelation lags and yields higher HAC variance estimates, whereas the shorter Newey–West bandwidth ($L = 5$) understates dependence and inflates SSR.

Despite this, cross-sectional ordering is essentially unchanged: the Spearman rank correlation between Andrews- and Newey–West-based SSR is 0.983, indicating near-perfect preservation of relative stability rankings. We therefore adopt Andrews' automatic bandwidth as our baseline, as it is data-driven and better suited to highly autocorrelated rolling-window statistics, while noting that our main ranking results are robust to this choice.

5.3.3 Subperiod analysis

To assess whether *SSR* rankings reflect persistent skill rather than regime-specific outcomes, we partition the full sample into three subperiods corresponding to distinct market regimes. Each subperiod reflects a different macro-financial backdrop in terms of volatility, monetary policy, and cross-asset co-movements.

Pre-GFC (January 1997 – June 2007). A global expansion phase characterized by moderate volatility, sustained equity bull markets, and conventional monetary policy. Ample liquidity, steady economic growth, and relatively stable ("normal") cross-asset correlation structures dominated this environment.

*GFC & Recovery (July 2007 – February 2020)***.** A regime shaped initially by the global financial crisis and subsequently by a prolonged recovery under ultra-accommodative monetary policy, including near-zero interest rates and large-scale quantitative easing. This period features episodes of severe market stress, major regime shifts such as the euro area sovereign debt crisis, and extended periods of compressed cash returns amid heavy central bank intervention.

Post-COVID (March 2020 – December 2025). A phase beginning with the pandemic-induced shock of 2020, marked by extreme initial volatility and unprecedented fiscal and monetary stimulus, followed by the 2022–2023 tightening cycle—the fastest rate-hiking episode in decades. This environment combines elevated inflation, sharp factor rotations, and structural market changes, including increased retail participation, the growing integration of crypto assets, and a more fragmented market microstructure.

TABLE 7
SSR Estimates Across Market Regimes

Strategy	Pre-GFC	GFC & Recovery	Post-COVID
Merger Arbitrage	1.03	1.10	1.33
Equity Long Short	1.12	0.80	1.83
Global Macro	2.60	0.97	4.31
European Equities	1.59	0.66	1.69
Event Driven	1.20	1.02	0.99
Hedge Fund	1.39	0.89	0.84
Technology	0.36	0.76	0.74
Healthcare & Biotechnology	0.71	0.68	0.36
Equity Long Bias	0.74	0.70	0.89
Convertible Arbitrage	0.70	0.82	0.91
Equity Market Neutral	2.81	0.45	1.94
Fixed Income Arbitrage	1.13	0.71	0.71
Pacific Rim Equities	0.84	0.60	0.96
Multi Strategy	1.87	0.73	1.36
Fund of Funds	2.47	0.34	0.69
Distressed Securities	0.94	0.49	0.95
Emerging Markets	0.61	0.44	0.36

Rank Correlation (Spearman):
Pre-GFC vs GFC-Recovery: 0.01 | Pre-GFC vs Post-COVID: 0.58 | GFC-Recovery vs Post-COVID: 0.34
Average: 0.31

Notes: Pre-GFC: Jan 1997–Jun 2007 (126 obs). GFC & Recovery: Jul 2007–Feb 2020 (152 obs). Post-COVID: Mar 2020–Dec 2025 (70 obs). *SSR* calculated with 36-month rolling window and Andrews bandwidth.

Table 7 reports *SSR*(0) for each subperiod. Cross-sectional rankings are markedly different across regimes: the average Spearman rank correlation is only 0.31, in sharp contrast to the much higher values obtained in the window-length (0.84) and bandwidth (0.98) sensitivity analyses. This contrast indicates that *SSR* is robust to reasonable methodological choices, and that the instability primarily reflects genuine regime dependence in hedge fund behavior rather than noise in the metric.

Both *SSR* levels and strategy rankings change substantially across regimes. Pre-GFC *SSRs* are generally high, with *Equity Market Neutral* (2.81) and *Global Macro* (2.60) at the top. During the GFC & Recovery period, *SSR* compresses across the board (average decline ≈45%) and the rank correlation with the Pre-GFC period collapses to 0.01; *Merger Arbitrage* (1.10) and *Event Driven* (1.02) then display the highest stability. Post-COVID, Global Macro jumps to 4.31—the highest *SSR* in the sample—while strategies such as *Healthcare & Biotechnology* and *Emerging Markets* fall to 0.36, consistent with their sensitivity to the pandemic–inflation–tightening sequence.

These patterns show that temporal stability, as measured by SSR, is inherently conditional on the prevailing market regime rather than an intrinsic, regime-invariant attribute of each strategy. *SSR* therefore captures how consistently a strategy performs *within* a given macro-financial environment and how its stability profile shifts *across* environments. The variation in rankings across subperiods is thus informative about long-run hedge fund behavior and regime robustness, not a limitation of the *SSR* metric itself.

5.3.4 Statistical Inference via Block Bootstrap

We employ a univariate moving block bootstrap procedure (Section 3.3.2) to construct finite-sample distributions of the SSR and to conduct formal inference on temporal stability. For each strategy, we generate B = 25,000 bootstrap replications using an optimal block length selected via the Politis–White (2004) method applied to the return series. Within each replication, we compute rolling 36-month Sharpe ratios using a benchmark $SR^* = 0$, HAC-robust standard deviations, and the corresponding SSR.

We test whether a strategy's temporal stability exceeds predefined thresholds $SSR^* \in \{0, 0.5\}$ by evaluating the one-sided hypotheses $H_0: SSR \leq SSR^*$against $H_1: SSR > SSR^*$. One-sided p-values are obtained as the proportion of bootstrap replications falling below each threshold. The null hypothesis is rejected at significance level $\alpha = 0.05$ when $\hat{p} < 0.05$, indicating statistically significant temporal stability. Percentile-based 95% confidence intervals are constructed from the empirical bootstrap quantiles.

Table 8 reports observed *SSR* estimates, confidence intervals, and p-values for the three stability thresholds. Figure 12 displays bootstrap distributions for three representative strategies with varying stability profiles.

TABLE 8
Bootstrap Statistical Inference Results

HF Strategy	SSR (0)	Bootstrap mean	95% CI Lower	95% CI Upper	p(SSR<0)	p(SSR<0.5)
Convertible Arbitrage	0.405	0.453	0.232	0.817	<0.001**	0.688
Distressed Securities	0.309	0.336	0.144	0.607	<0.001**	0.909
Emerging Markets	0.269	0.278	0.074	0.559	<0.001**	0.950
Equity Long Bias	0.423	0.402	0.188	0.717	<0.001**	0.795
Equity Long/Short	0.573	0.582	0.312	1.003	<0.001**	0.357
Equity Market Neutral	0.404	0.780	0.458	1.282	<0.001**	0.055
European Equities	0.492	0.553	0.287	0.971	<0.001**	0.434
Event Driven	0.466	0.434	0.231	0.743	<0.001**	0.737
Fixed Income Arbitrage	0.402	0.433	0.223	0.790	<0.001**	0.743
Fund of Funds	0.319	0.373	0.150	0.708	<0.001**	0.833
Global Macro	0.568	0.716	0.403	1.204	<0.001**	0.122
Healthcare & Biotechnology	0.430	0.426	0.195	0.760	<0.001**	0.735
Hedge Fund	0.457	0.465	0.231	0.823	<0.001**	0.649
Merger Arbitrage	0.621	0.680	0.410	1.150	<0.001**	0.145
Multi-Strategy	0.364	0.479	0.265	0.831	<0.001**	0.629
Pacific Rim Equities	0.378	0.344	0.149	0.618	<0.001**	0.902
Technology	0.454	0.437	0.204	0.779	<0.001**	0.714

Notes: ** denote rejection of H_0 at the 1% significance level.

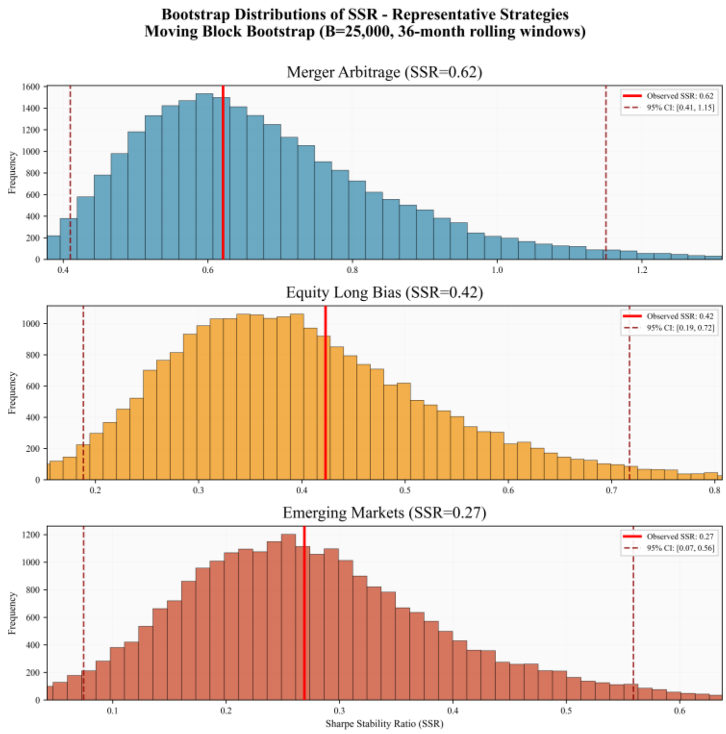
Bootstrap Inference Results

All 17 hedge fund strategies decisively reject the null hypothesis of zero temporal stability ($p < 0.01$), confirming that positive *SSR* estimates reflect genuine temporal consistency rather than sampling noise. This result indicates that risk-adjusted returns in hedge fund strategy indices exhibit statistically detectable stability once serial dependence and overlapping observations are properly accounted for through HAC-robust block-bootstrap inference.

At the more demanding stability threshold (SSR > 0.5), evidence weakens substantially. No strategy rejects the null at the 5% level, although Equity Market Neutral ($p = 0.055$), Global Macro ($p = 0.122$), and Merger Arbitrage ($p = 0.145$) approach marginal significance. For the remaining strategies, positive point estimates of *SSR* remain statistically indistinguishable from moderate temporal stability after conservative finite-sample adjustment.

Bootstrap confidence intervals are correspondingly wide (Figure 12), reflecting limited effective sample sizes induced by strong serial dependence in rolling Sharpe ratios rather than methodological shortcomings. For example, Merger Arbitrage's 95% interval [0.410, 1.150] spans a broad range of stability outcomes. This conservatism is a deliberate feature of the block-bootstrap framework, which penalizes inflated precision arising from overlapping windows and avoids overconfidence implicit in IID-based inference.

FIGURE 12
Bootstrap Distributions of *SSR* for three representative strategies



Overall, these results highlight SSR's complementary role relative to static performance metrics. While most strategies exhibit statistically credible average performance (high *SR* and *PSR* relative to $SR^* = 0$), fewer than 20% demonstrate temporal stability exceeding moderate thresholds with statistical confidence. This divergence underscores that strong average risk-adjusted returns need not imply persistent performance: episodic outperformance can generate high *SR* without sustained stability. *SSR* therefore provides an economically meaningful lens for distinguishing robustness from intermittency—an essential consideration for institutional investors assessing manager skill and portfolio resilience across time.

6. CONCLUSION

This paper introduces the *Sharpe Stability Ratio* (*SSR*), a performance metric that evaluates the temporal consistency of risk-adjusted returns. *SSR* extends point-in-time credibility assessment to temporal inference by applying heteroskedasticity- and autocorrelation-robust (HAC) variance estimation to rolling Sharpe ratio sequences, addressing a fundamental gap: while the Sharpe ratio (*SR*) measures aggregate performance and the Probabilistic Sharpe Ratio (*PSR*) evaluates its statistical credibility under non-Normal returns, neither quantifies whether credible performance reflects persistent skill or episodic outperformance.

Key findings are as follows. First, strategies with identical ex-post *SR* and *PSR* can display markedly different temporal stability profiles, as synthetic portfolios calibrated to the same full-sample *SR* show that *SSR* captures information orthogonal to static metrics. Second, overlapping rolling windows generate strong serial dependence in *SR* time series, so naive variance estimators understate uncertainty by orders of magnitude, making HAC corrections and dependence-preserving block bootstrap procedures essential for valid temporal stability inference. Third, statistically credible aggregate performance does not guarantee temporal consistency: analysis of hedge fund strategy indices indicates that although most exhibit statistically credible point-in-time performance (high *PSR*), substantially fewer exceed moderate *SSR* thresholds, revealing that static credibility can mask episodic concentration of returns. Fourth, temporal stability emerges as a regime-dependent property rather than a fixed, intrinsic characteristic of each strategy: *SSR* rankings shift substantially across pre-GFC, GFC-and-recovery, and post-COVID periods, reflecting genuine differences in how hedge funds respond to changing macro-financial environments rather than instability of the metric itself.

From a practical perspective, *SSR* provides actionable information for due diligence and ongoing monitoring. Investors can combine *PSR* (credibility) with *SSR* (consistency) to separate persistent skill from episodic outperformance using only standard inputs—returns, rolling windows, and HAC variance estimates—so implementation is straightforward and does not require additional data. As an extension, we introduce the Credibility-Adjusted SSR (CASSR), which integrates temporal stability with point-in-time credibility into a unified metric, though its full theoretical analysis is left for future research.

Several limitations qualify these contributions. *SSR* relies on approximate stationarity of returns and rolling statistics, so structural breaks or sharp regime shifts can lower *SSR* even when instability mainly reflects regime heterogeneity; in addition, the choice of rolling window length, HAC bandwidth, and bootstrap block length entails trade-offs between statistical precision and responsiveness to changing conditions. Moreover, hedge fund data are subject to survivorship bias, backfill, and return smoothing, which implies that *SSR* should be interpreted as a diagnostic complement rather than a substitute for existing performance measures.

Future research could develop regime-conditional *SSR* decompositions that distinguish stability across market environments, integrate *SSR* with the Deflated Sharpe Ratio to jointly address multiple-testing bias and temporal consistency, further analyze the proposed *CASSR*, and link *SSR* to underlying sources of performance through factor-attribution frameworks.

SSR thus offers a statistically grounded method to incorporate temporal stability into performance evaluation by quantifying how much of a strategy's Sharpe ratio reflects persistent performance versus concentrated episodic strength, capturing a dimension of performance measurement that point-in-time metrics inherently miss.

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APPENDIX A. TECHNICAL NOTES

A.1 MINIMUM TRACK RECORD LENGTH

PSR provides a probabilistic assessment of whether an observed Sharpe ratio $\widehat{SR}$ exceeds a benchmark SR^*. For a confidence level $1 - \alpha$, and a return distribution characterized by skewness γ_3 and kurtosis γ_4, the Minimum Track Record Length (*MinTRL*) is given by

$$MinTRL(\widehat{SR}, SR^*, \alpha; \hat{\gamma}_3, \hat{\gamma}_4) = 1 + \left[1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2\right] \left(\frac{Z_\alpha}{\widehat{SR} - SR^*}\right)^2,$$

where Z_α is the standard normal quantile (e.g., $Z_{0.05} \approx 1.645$ for one-sided 95% confidence).

MinTRL represents the minimum number of observations required for the estimated Sharpe ratio to exceed the benchmark SR^* with probability at least $1 - \alpha$ under the asymptotic normal approximation underlying *PSR*.

The expression implies:

(i) Higher $(\widehat{SR} - SR^*)$ implies shorter *MinTRL* (stronger performance is easier to detect).
(ii) Higher confidence $1 - \alpha$ increases *MinTRL*.
(iii) Negative skewness and excess kurtosis increase *MinTRL*, reflecting higher estimation uncertainty under non-Normal returns.

If returns are observed at frequency q, *MinTRL* expressed in calendar time equals $MinTRL/q$years.

Table A.1 reports MinTRL values (in daily observations and equivalent years) assuming $q = 252$ trading days per year and $\alpha = 0.05$. Annualized Sharpe ratios are converted to daily frequency via $SR_{\text{daily}} = SR_{\text{ann}}/\sqrt{252}$.

Table A.1 *Minimum Track Record Length (daily observations and years)*

Table A.1 shows the *MinTRL* required to reject H_0: $SR \leq 0$ at 95% confidence level for various observed annualized Sharpe ratios and return distributions:

***SR* (ann.)**	**IID Normal (γ_3=0, γ_4=3)**	**Negative skew (γ_3=-0.5, γ_4=5)**	**Fat tails (γ_3=-1, γ_4=8)**
0.5	2729 obs (10.8 years)	2773 obs (11.0 years)	2818 obs (11.2 years)
1.0	684 obs (2.7 years)	706 obs (2.8 years)	730 obs (2.9 years)
1.5	305 obs (1.2 years)	321 obs (1.3 years)	337 obs (1.3 years)
2.0	172 obs (0.7 years)	184 obs (0.7 years)	197 obs (0.8 years)

Notes: *Annualized Sharpe ratios converted to daily using $SR_{daily} = SR_{ann}/\sqrt{252}$. MinTRL computed for SR^* = 0 with α = 0.05. Negative skewness and fat tails increase MinTRL by 1.6%-14.5% relative to IID Normal returns.*

MinTRL values should be interpreted as asymptotic guidelines rather than exact finite-sample thresholds. For small samples or low Sharpe ratios, the normal approximation underlying *PSR* may introduce finite-sample error, so *MinTRL* provides a conservative benchmark rather than a sharp cutoff.

A.2 AUTOCORRELATION FROM OVERLAPPING WINDOWS

To understand the autocorrelation induced by overlapping windows, we begin with the simplest case: rolling means of excess returns. This provides a theoretical benchmark against which to evaluate the more complex rolling *SR* series.

Consider excess returns $\{r_t\}$ with mean zero, variance σ_r^2, and weak dependence as in Assumption 3.1. For a given window length τ, define the rolling mean

$$\bar{r}_{t,\tau} = \frac{1}{\tau}\sum_{j=0}^{\tau-1} r_{t-j}\,, t = \tau, \ldots, T.$$

Proposition A.1 *(Autocorrelation of Linear Rolling Statistics).* Under IID returns with finite second moments, the lag-*k* autocorrelation of rolling means satisfies:

$$\rho_k \approx \begin{cases} 1 - \dfrac{k}{\tau}, & \text{for } k < \tau, \\ 0, & \text{for } k \geq \tau. \end{cases}$$

In particular, the first-order autocorrelation is $\rho_1 \approx \frac{\tau-1}{\tau}$.

Proof.

For $k < \tau$, the rolling windows share $\tau - k$ observations. Under independence,

$$\text{Cov}(\bar{r}_{t,\tau}, \bar{r}_{t-k,\tau}) = \frac{1}{\tau^2}\sum_{s=0}^{\tau-k-1} \text{Var}(r_{t-s}) = \frac{\tau - k}{\tau^2}\sigma_r^2.$$

The variance of the rolling mean is

$$\text{Var}(\bar{r}_{t,\tau}) = \frac{1}{\tau^2}\sum_{s=0}^{\tau-1} \text{Var}(r_{t-s}) = \frac{\sigma_r^2}{\tau}.$$

Therefore,

$$\rho_k = \frac{\text{Cov}(\bar{r}_{t,\tau}, \bar{r}_{t-k,\tau})}{\text{Var}(\bar{r}_{t,\tau})} = \frac{(\tau - k)\sigma_r^2/\tau^2}{\sigma_r^2/\tau} = 1 - \frac{k}{\tau}.$$

For $k \geq \tau$, the windows do not overlap and the covariance is zero, hence $\rho_k = 0$. ■

Interpretation. This result shows that mechanical overlap alone—without any serial correlation in the underlying returns—creates strong positive autocorrelation in rolling statistics. For typical parameter values:

- Daily data with annual window (τ = 252): $\rho_1 \approx 0.996$
- Monthly data with 3-year window (τ = 36): $\rho_1 \approx 0.972$

Thus, rolling statistics constructed from overlapping windows are near-unit-root processes for the first $\tau - 1$ lags, with autocorrelation decaying linearly to zero at lag τ. Importantly, this dependence is purely mechanical and does not reflect persistence in the underlying return process. By delta-method arguments, nonlinear rolling statistics such as rolling standard deviations and rolling Sharpe ratios inherit this autocorrelation structure to first order, since they are smooth transformations of rolling means and rolling second moments (see Remark 3.1).

A.3 VARIANCE UNDERSTATEMENT UNDER OVERLAPPING WINDOWS

Practitioners often compute the sample variance of the rolling Sharpe process Z_t as

$$\hat{\gamma}_0 = \frac{1}{N-1} \sum_{t=1}^{N} (Z_t - \bar{Z})^2.$$

This estimator is consistent for the instantaneous variance $\gamma_0 = \mathrm{Var}(Z_t)$, but it does not estimate the long-run variance

$$\sigma_\infty^2 = \gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k,$$

which governs the sampling distribution of the sample mean under serial dependence. When rolling windows overlap, positive autocorrelation inflates the long-run variance relative to the instantaneous variance.

Proposition A.2 *(Order of Variance Understatement Under Overlap)*

Under the overlap-induced autocorrelation pattern

$$\rho_k \approx \begin{cases} 1 - \dfrac{k}{\tau}, & k < \tau, \\ 0, & k \geq \tau, \end{cases}$$

the long-run variance satisfies

$$\frac{\sigma_\infty^2}{\gamma_0} \approx \tau.$$

Thus, to first order in τ, the naive variance estimator understates the long-run variance by a factor of order τ.

Proof.

By definition,

$$\sigma_\infty^2 = \gamma_0 + 2\sum_{k=1}^{\infty} \gamma_k.$$

Writing $\gamma_k = \rho_k \gamma_0$, we obtain

$$\sigma_\infty^2 = \gamma_0 (1+2\textstyle\sum_{k=1}^{\infty} \rho_k).$$

Under the overlap structure,

$$\sum_{k=1}^{\infty} \rho_k = \sum_{k=1}^{\tau-1} \left(1 - \frac{k}{\tau}\right).$$

Compute the sum:

$$\sum_{k=1}^{\tau-1} \left(1 - \frac{k}{\tau}\right) = (\tau - 1) - \frac{1}{\tau}\sum_{k=1}^{\tau-1} k.$$

Since

$$\sum_{k=1}^{\tau-1} k = \frac{(\tau-1)\tau}{2},$$

we obtain

$$(\tau - 1) - \frac{1}{\tau} \cdot \frac{(\tau-1)\tau}{2} = \frac{\tau - 1}{2}.$$

Therefore,

$$\sigma_\infty^2 \approx \gamma_0 \left(1 + 2 \cdot \frac{\tau - 1}{2}\right) = \gamma_0 \tau.$$

Hence

$$\frac{\sigma_\infty^2}{\gamma_0} \approx \tau.$$

■

Consequence for Standard Errors

$$\frac{\hat{\sigma}_{\text{naive}}}{\hat{\sigma}_{\text{HAC}}} = \sqrt{\frac{\hat{\gamma}_0}{\hat{\sigma}_\infty^2}} \approx \frac{1}{\sqrt{\tau}}.$$

Thus, naive standard errors shrink by a factor of order $\sqrt{\tau}$, leading to severe overstatement of statistical significance.

Numerical Magnitude. For typical rolling window lengths:

- Monthly data (τ = 36): Naive variance understates by a factor of order 36
- Daily data (τ = 252): Naive variance understates by a factor of order 252

Thus, naive standard errors shrink by a factor of order $\sqrt{\tau}$, leading to severe overstatement of statistical significance.

A.4 HETEROSKEDASTICITY AND AUTOCORRELATION CONSISTENT (HAC) VARIANCE ESTIMATION

The mechanical autocorrelation documented in Proposition 3.1 has immediate implications for statistical inference on the mean of the rolling performance process Z_t. Throughout this appendix, let $\{Z_t\}_{t=1}^N$ denote the rolling Sharpe process defined in Section 3.2.

Standard inference on sample means relies on the Central Limit Theorem. For a strictly stationary process $\{Z_t\}$ with mean μ_Z and finite second moments,

$$\sqrt{N}(\bar{Z} - \mu_Z) \xrightarrow{d} \mathcal{N}(0, \sigma_\infty^2),$$

where the long-run variance is

$$\sigma_\infty^2 = \sum_{k=-\infty}^{\infty} \gamma_k = \gamma_0 + 2\sum_{k=1}^{\infty} \gamma_k$$

with $\gamma_k = \text{Cov}(Z_t, Z_{t-k})$. The long-run variance captures the cumulative impact of serial dependence on the sampling distribution of $\bar{Z}$.

Naive Variance and Its Failure

The sample variance

$$\hat{\sigma}_{\text{naive}}^2 = \frac{1}{N-1}\sum_{t=1}^{N}(Z_t - \bar{Z})^2$$

consistently estimates γ_0, but not σ_∞^2. When Z_t exhibits positive autocorrelation — as induced mechanically by overlapping windows — ignoring the autocovariance terms $2\sum_{k=1}^{\infty}\gamma_k$ leads to systematic understatement of the true variance of $\bar{Z}$. As shown in Appendix A.3, this understatement can be of order τ for rolling statistics.

A.4.1 HAC estimation

To obtain valid inference on $\bar{Z}$ under serial correlation, we require a heteroskedasticity- and autocorrelation-consistent estimator of σ_∞^2. The Newey–West (1987) estimator with Bartlett kernel is

$$\hat{\sigma}_{\text{HAC}}^2 = \hat{\gamma}_0 + 2\sum_{k=1}^{L} \kappa\left(\frac{k}{L+1}\right)\hat{\gamma}_k$$

where

$$\hat{\gamma}_k = \frac{1}{N}\sum_{t=k+1}^{N}(Z_t - \bar{Z})(Z_{t-k} - \bar{Z}), k \geq 1,$$

and

$$\kappa(z) = 1 - | z |, | z | \leq 1,$$

so that

$$\kappa\left(\frac{k}{L+1}\right) = 1 - \frac{k}{L+1}.$$

Here:

- $\hat{\gamma}_0$ is the sample variance,
- L is the bandwidth parameter,
- the kernel truncates autocovariances beyond lag L.

A.4.2 Bartlett kernel

Alternative kernels (Parzen, Quadratic Spectral) provide modest asymptotic efficiency gains (Andrews, 1991). However, the Bartlett kernel guarantees positive semi-definiteness of the estimator and exhibits robust finite-sample behavior for highly persistent series such as rolling Sharpe ratios.

The Bartlett kernel applies linearly declining weights of the form

$$\kappa\left(\frac{k}{L+1}\right) = 1 - \frac{k}{L+1}, k = 0, \ldots, L,$$

so that low-order autocovariances receive weight close to one, while higher-order lags are progressively downweighted and truncated at lag L.

This weighting serves two purposes:

1. *Variance control.* Sample autocovariances at higher lags are estimated with greater sampling variability, since fewer overlapping pairs are available and the true covariance is typically smaller. Downweighting them reduces estimator variance.
2. *Bias control under weak dependence.* For stationary processes with absolutely summable autocovariances, $\gamma_k \to 0$ as $k \to \infty$. Assigning smaller weights to distant lags ensures that the estimator remains consistent while limiting the impact of noisy high-lag terms.

The triangular weighting thus provides a simple, positive semi-definite estimator that balances bias and variance under general weak dependence.

A.4.3 Bandwidth selection

The bandwidth Ltrades off bias and variance:

- Small L omits relevant autocovariances (downward bias).
- Large L includes noisy high-lag estimates (variance inflation).

Newey and West (1987) proposed the rule

$$L_{NW} = \lfloor 4\left(N/100\right)^{2/9} \rfloor.$$

This rule was derived under moderate serial dependence and may be inappropriate for rolling Sharpe processes, whose mechanical overlap induces near-unit autocorrelation for the first $\tau - 1$ lags.

We therefore employ Andrews' (1991) automatic bandwidth selection, which minimizes asymptotic mean squared error of the HAC estimator by estimating the spectral density at frequency zero. This data-dependent procedure adapts to the actual persistence of each rolling Sharpe sequence.

A.4.4 Consistency

Under Assumption 3.1 and standard regularity conditions,

$$\hat{\sigma}^2_{\mathrm{HAC}} \xrightarrow{p} \sigma^2_{\infty} \text{as } N \to \infty,$$

provided

$$L \to \infty, \frac{L}{N} \to 0.$$

(Newey and West, 1987; Andrews, 1991). These conditions ensure asymptotically valid inference for the benchmark-centered Sharpe Stability statistic developed in Section 3.3.

APPENDIX B. TECHNICAL ASSUMPTIONS AND ASYMTOTIC PROOFS

B.1 Mixing and Moment Conditions

Let $\{r_t\}$ denote the excess-return process and $\{Z_t\}_{t=1}^{N}$ the rolling Sharpe process defined in Section 3.2.

Assumption B.1 *(Strong mixing of returns)*

The excess-return process $\{r_t\}$ is assumed strictly stationary and α-mixing with summable coefficients $\sum_{k=1}^{\infty} \alpha(k) < \infty$, where the α-mixing coefficient is defined as:

$$\alpha(k) = \sup_{A \in \mathcal{F}_{-\infty}^{t}, B \in \mathcal{F}_{t+k}^{\infty}} \mid P(A \cap B) - P(A)P(B) \mid$$

and $\mathcal{F}_a^b$ denotes the σ-algebra generated by $\{r_s : a \leq s \leq b\}$. We assume polynomial decay

$$\alpha(k) = O\left(k^{-\delta}\right) \text{for some } \delta > 2.$$

Assumption B.2 *(Moment condition)*

There exists $\varepsilon > 0$ such that

$$E \mid Z_t \mid^{2+\varepsilon} < \infty.$$

Assumption B.3 *(Summability of autocovariances)*

The rolling Sharpe process satisfies

$$\sum_{k=1}^{\infty} \mid \gamma_k \mid < \infty, \gamma_k = \mathrm{Cov}(Z_t, Z_{t-k}).$$

Assumption B.3 ensures that the long-run variance

$$\sigma_\infty^2 = \gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k$$

exists and is finite.

Remarks:

- Under smooth functional transformations, rolling statistics Z_t inherit the mixing properties of $\{r_t\}$ (near-epoch dependence).
- Standard ARMA, GARCH, and Markov-switching processes satisfy these conditions under well-known parameter restrictions (Abramson and Cohen, 2007).

B.2 Consistency of HAC Estimator

Assumption B.4 *(Bandwidth growth)*

Let $L = L_N$ denote the HAC bandwidth. We require

$$L \to \infty, \frac{L}{N} \to 0 \text{ as } N \to \infty.$$

Theorem 3.1 *(HAC Consistency)*

Under Assumptions B.1–B.4,

$$\hat{\sigma}^2_{\text{HAC}} = \hat{\gamma}_0 + 2 \sum_{k=1}^{L} \kappa \left(\frac{k}{L+1}\right) \hat{\gamma}_k \xrightarrow{p} \sigma^2_\infty.$$

Proof.

Under α-mixing with polynomial decay and finite $2 + \varepsilon$ moments:

1. Sample autocovariances satisfy $\hat{\gamma}_k \to_p \gamma_k$ for fixed k.
2. Absolute summability of γ_k ensures truncation bias vanishes as $L \to \infty$.
3. The condition $L/N \to 0$ controls estimator variance.
4. The Bartlett kernel is bounded and continuous at zero.

Therefore the Newey–West estimator is consistent (Newey and West, 1987; Andrews, 1991). ■

Remark B.1 *(Rate)*

If $L = O(N^{1/3})$, then

$$\hat{\sigma}^2_{\text{HAC}} - \sigma^2_\infty = O_p(N^{-1/3}),$$

which is slower than the parametric $N^{-1/2}$ rate (Andrews, 1991). This affects convergence speed but not asymptotic validity.

B.3 Central Limit Theorem for Rolling Sharpe Mean

Theorem 3.2

Under Assumptions B.1–B.3,

$$\sqrt{N}(\bar{Z} - \mu_Z) \xrightarrow{d} \mathcal{N}(0, \sigma_\infty^2).$$

Proof.

Under α-mixing with polynomial decay $\delta > 2$ and finite $2 + \varepsilon$ moments, the central limit theorem for strongly mixing processes applies (Ibragimov, 1962; Billingsley, 1968).

Absolute summability of autocovariances ensures the asymptotic variance equals σ_∞^2. ■

B.4 Asymptotic Distribution of the *SSR* Statistic

Define the benchmark-centered estimator

$$\widehat{SSR}(SR^*) = \frac{\bar{Z} - SR^*}{\hat{\sigma}_{\text{HAC}}}.$$

Let

$$t_{SSR}(SR^*) = \sqrt{N}\,\widehat{SSR}(SR^*).$$

Theorem 3.3

Under Assumptions B.1–B.4,

$$t_{SSR}(SR^*) = \frac{\sqrt{N}(\bar{Z} - SR^*)}{\hat{\sigma}_{\text{HAC}}} \xrightarrow{d} \mathcal{N}(0,1) \text{ under } H_0\colon \mu_Z = SR^*.$$

Proof

From Theorem 3.2,

$$\sqrt{N}(\bar{Z} - \mu_Z) \xrightarrow{d} \mathcal{N}(0, \sigma_\infty^2).$$

Under $H_0\colon \mu_Z = SR^*$,

$$\sqrt{N}(\bar{Z} - SR^*) = \sqrt{N}(\bar{Z} - \mu_Z).$$

From Theorem 3.1,

$$\hat{\sigma}_{\text{HAC}} \xrightarrow{p} \sigma_\infty.$$

By Slutsky's theorem,

$$\frac{\sqrt{N}(\bar{Z} - SR^*)}{\hat{\sigma}_{\text{HAC}}} \xrightarrow{d} \mathcal{N}(0,1).$$

■

APPENDIX C

BOOTSTRAP REFINEMENT FOR FINITE-SAMPLE INFERENCE

The asymptotic theory in Section 3 establishes that the benchmark-centered Sharpe Stability Ratio statistic

$$t_{SSR}(SR^*) = \frac{\sqrt{N}(\bar{Z} - SR^*)}{\hat{\sigma}_{HAC}(Z)}$$

is asymptotically standard normal under the stated mixing and moment conditions. This provides the primary inferential foundation of the paper.

Rolling performance statistics constructed from overlapping windows exhibit strong mechanical persistence, which may generate finite-sample distortions despite asymptotic validity. The moving block bootstrap (MBB) is therefore implemented as a finite-sample refinement of the asymptotic inference (Künsch, 1989; Hall et al., 1995). It does not replace the asymptotic framework, but approximates the sampling distribution of the SSR statistic under the observed dependence structure.

Bootstrap resampling is applied to the return series, with rolling Sharpe ratios and SSR reconstructed within each replication (returns → rolling SR → SSR). This preserves both genuine serial dependence in returns and overlap-induced autocorrelation in rolling statistics.

C.1 Moving Block Bootstrap Procedure

Let $\{r_t\}_{t=1}^{T}$ denote the excess return series. Let τ denote the rolling window length and $N = T - \tau + 1$ the number of rolling observations.

Block length ℓ is selected using the automatic data-dependent procedure of Politis and White (2004), as corrected by Patton et al. (2009). For asymptotic validity of the MBB, the block length satisfies

$$\ell \to \infty, \frac{\ell}{T} \to 0 \text{ as } T \to \infty$$

(Künsch, 1989; Hall et al., 1995).

Algorithm C.1: Moving Block Bootstrap for *SSR*

Inputs:

Returns $\{r_t\}_{t=1}^{T}$, rolling window τ, benchmark SR^*, bootstrap replications B, block length ℓ.

Output:

Bootstrap sample $\{\widehat{SSR}^{(b)}(SR^*)\}_{b=1}^{B}$.

1. Construct overlapping blocks

$$B_j = (r_j, \dots, r_{j+\ell-1}), j = 1, \dots, T - \ell + 1.$$

2. For $b = 1, \dots, B$:

(a) Draw $K = \lceil T/\ell \rceil$ blocks independently with replacement from $\{B_j\}$.

(b) Concatenate selected blocks and truncate to length T to obtain bootstrap returns $\{r_t^{(b)}\}_{t=1}^{T}$.

(c) Compute rolling Sharpe sequence $\{Z_t^{(b)}\}_{t=1}^{N}$ using window τ.

(d) Compute

$$\bar{Z}^{(b)} = \frac{1}{N}\sum_{t=1}^{N} Z_t^{(b)}.$$

(e) Compute HAC standard deviation

$$\hat{\sigma}_{HAC}^{(b)} = \hat{\sigma}_{HAC}(\{Z_t^{(b)}\}_{t=1}^{N}),$$

using the same kernel and bandwidth rule as in the original sample.

(f) Form

$$\widehat{SSR}^{(b)}(SR^*) = \frac{\bar{Z}^{(b)} - SR^*}{\hat{\sigma}_{HAC}^{(b)}}.$$

The empirical distribution $\{\widehat{SSR}^{(b)}(SR^*)\}_{b=1}^{B}$ approximates the finite-sample distribution of the centered SSR statistic under the dependence structure of the observed return process.

C.2 Bootstrap-Refined Hypothesis Testing

Inference remains grounded in the asymptotic normality result of Theorem 3.3. Bootstrap procedures provide finite-sample refinement when persistence is strong.

C.2.1 Stability vs. Benchmark

We test

$$H_0: \mu_Z \leq SR^* \text{vs.} H_1: \mu_Z > SR^*.$$

Let

$$SSR_{obs} = \frac{\bar{Z} - SR^*}{\hat{\sigma}_{HAC}}.$$

The one-sided bootstrap p-value is

$$\hat{p} = \frac{1}{B} \sum_{b=1}^{B} 1\{\widehat{SSR}^{(b)}(SR^*) \leq 0\}.$$

This tail probability provides a finite-sample approximation to the sampling distribution of the centered SSR statistic, complementing the asymptotic normal benchmark.

Percentile confidence intervals are constructed from the empirical quantiles

$$CI_{1-\alpha} = [q_{\alpha/2}, q_{1-\alpha/2}].$$

Rejection at level α occurs when the lower bound exceeds zero.

C.2.2 Threshold Stability Tests

For a stability threshold θ, we test

$$H_0: SSR(SR^*) \leq \theta \text{ vs. } H_1: SSR(SR^*) > \theta.$$

The bootstrap p-value is

$$\hat{p} = \frac{1}{B} \sum_{b=1}^{B} 1\{\widehat{SSR}^{(b)}(SR^*) < \theta\}.$$

Interpretation follows analogously as a finite-sample refinement of the asymptotic test.

C.2.3 Pairwise Stability Comparisons

For two strategies A and B, blocks are drawn jointly to preserve cross-sectional dependence. For each replication:

$$\widehat{SSR}_A^{(b)} = \frac{\bar{Z}_A^{(b)} - SR^*}{\hat{\sigma}_{HAC,A}^{(b)}}, \widehat{SSR}_B^{(b)} = \frac{\bar{Z}_B^{(b)} - SR^*}{\hat{\sigma}_{HAC,B}^{(b)}}.$$

Define

$$\Delta\widehat{SSR}^{(b)} = \widehat{SSR}_A^{(b)} - \widehat{SSR}_B^{(b)}.$$

The one-sided p-value is

$$\hat{p} = \frac{1}{B}\sum_{b=1}^{B} 1\{\Delta\,\widehat{SSR}^{(b)} < 0\}.$$

These p-values are marginal measures of evidence. When multiple comparisons are conducted, standard multiple-testing adjustments may be applied.